Project Portfolio Budgeting as a Markov Decision Process: Environment Design, Baseline Benchmarks, and Optimality Gap Analysis

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Abstract

Allocating a shared capital reserve across a portfolio of simultaneously executing contracted projects is a high-frequency, high-stakes decision for engineering contractors. Classical approaches either assume full foresight or collapse the portfolio to a single project, leaving practitioners with no principled tool for the sequential, multi-project setting under real-world uncertainty. This paper closes that gap by delivering a complete, verified simulation environment that makes the problem tractable for data-driven methods.

We formalise the problem as a Constrained Markov Decision Process in which a portfolio manager allocates budget fractions across N concurrent projects at each discrete period. The state encodes Earned Value Management indicators, S-curve planned progress, milestone payment lookaheads, and a per-project breach-tolerance counter. The transition dynamics reproduce the full FIDIC contract cash-flow cycle: advance payment and recovery, progress-weighted milestone gross payments, retention withholding and release, interest charges on treasury draws, and a two-condition termination gate. The reward signal is the γ-discounted portfolio net cash flow, accumulating to an episode NPV that serves as the primary performance criterion.

The environment is released as a configurable, Gymnasium-compatible package with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag — eliminating the black-box problem inherent to learned policies. To establish a performance baseline and motivate the need for learned agents, we run three deterministic greedy heuristics (Reach-Plan, Catchup-T, and Catchup-Next-T) and a manually computed full-information optimum across four benchmark configurations of increasing portfolio complexity. The results show a systematic and widening gap between the greedy baselines and the optimal solution as portfolio complexity grows, confirming that standard rule-based allocation is insufficient and that a well-defined improvement target exists for reinforcement learning agents. Training such agents and evaluating them against a formal Linear Programming bound are identified as the natural next steps, addressed in subsequent work.

Keywords: project portfolio management; budget allocation under performance uncertainty; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; Gymnasium environment; greedy baseline; optimality gap.

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1 Introduction

1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules (Flyvbjerg et al., 2003; Merrow, 2011). When a portfolio of such projects executes simultaneously under a shared capital commitment, the risks compound: a performance shortfall on one project propagates through the shared budget to all others, and the decision of how much to allocate to each project at each period can determine whether the portfolio succeeds or fails.

Despite this, the formal treatment of portfolio-level budget allocation as a sequential decision problem is absent from the project portfolio management literature. Research has concentrated on the *selection* stage — which projects to include — while the *operational governance* stage — how to manage a committed portfolio through to completion — has received comparatively little analytical treatment (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Turner, 2014). The present work fills this gap by delivering a verified simulation environment that makes the problem tractable for data-driven methods and establishing a performance baseline that quantifies the cost of naive allocation.

We consider a portfolio of n projects, each governed by a bilateral contract consistent with FIDIC forms (Fédération Internationale des Ingénieurs-Conseils, 2017), executing concurrently under a relaxed capital commitment. A portfolio manager exercises one recurring decision at each discrete planning period: how much budget to allocate to each active project. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while respecting contract mechanics.

1.2 The Project Management Decision Hierarchy

The present contribution operates at the portfolio governance layer of the three-tier project management hierarchy (Project Management Institute, 2017). Portfolio selection — which projects to commit to — is treated as an exogenous input, consistent with the two-stage governance model of Turner (2014). At the governance layer, the portfolio manager does not control project execution directly but allocates budget to each project and observes earned value outcomes. Termination is not a deliberate agent action but an endogenous environment mechanic. It is triggered by one of two independent conditions. The first is a hard deadline: if the current episode timestep reaches planned finish + finish delay cap, the project is terminated immediately regardless of progress or tolerance. The second is sustained underperformance: if a projects actual progress falls behind its planned progress by more than progress delay cap, a tolerance counter decrements by one each period the breach persists and resets to its initial value whenever the breach clears; termination fires when the counter reaches zero.

Project termination differs from episode termination a project terminates individually and triggers a settlement cash flow, while the episode ends only when every project has either completed or been terminated, or when the global horizon is reached, or when the portfolio budget is exhausted with no pending inflows.

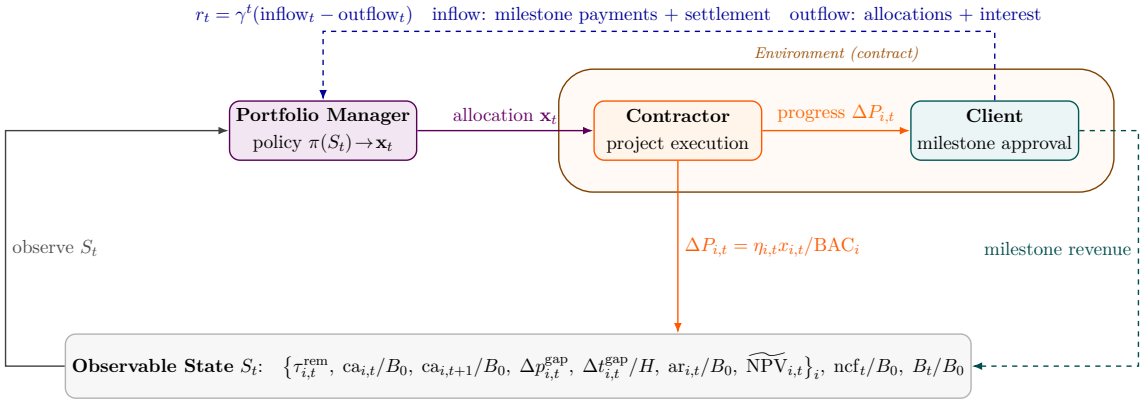


Figure 1: Portfolio governance as a reinforcement learning feedback loop. B_0: initial budget; H: horizon; τ^{rem}: tolerance remaining; ca: catchup allocation; Δp^{gap}: progress gap to next milestone; Δt^{gap}: timestep gap to milestone deadline; ar: required allocation to reach milestone; $\widetilde{\text{NPV}}$: normalised discounted milestone value; ncf: net cashflow.

1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager and each contractor is governed by FIDIC contract forms (Fédération Internationale des Ingénieurs-Conseils, 2017). Three mechanisms determine the cash-flow structure.

Milestone-based payment. Payment is contingent on certified progress thresholds. The gross payment at milestone j is a contractually fixed fraction $\phi_{i,j}$ of the contract price $CP_i = (1 + \pi_i) \cdot \text{BAC}_i$. Because payment is contingent on certified progress, the timing of cash inflows depends on the contractor's actual performance and the portfolio manager's allocation decisions. Cash-flow uncertainty in the present model arises from a single source: the stochastic budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period. Uncertain productivity induces uncertain milestone timing, which in turn induces uncertain payment arrival — without requiring any separate model of payment delay. This formulation is sufficient to justify the uncertainty framing of the title and admits a cleaner identification of the allocation problem structure than models with multiple independent uncertainty sources.

Advance payment and progressive recovery. At project initiation, an advance payment $A_i = \alpha_i \cdot CP_i$ is disbursed to the contractor. It is recovered through percentage deductions from subsequent interim payment certificates, with commencement threshold ψ_i and deduction rate δ_i^{rec} specified as project-specific contract parameters (Fédération Internationale des Ingénieurs-Conseils, 2017).

Retention. A fraction ρ_i of each milestone payment is withheld as a performance bond. The accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released in full upon project completion, creating a structural completion incentive embedded in the payment mechanics.

Earned value management as a performance signal. The EVM framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary for monitoring contractor performance. Define the Budgeted Cost of Work Performed as $\text{BCWP}_{i,t} = \text{EV}_{i,t} \cdot \text{BAC}_i$, where $\text{EV}_{i,t} \in [0, 1]$ is the fraction of work completed, and the Actual

Cost of Work Performed as $\mathrm{ACWP}_{i,t} = \sum_{\tau \le t} x_{i,\tau}$, the cumulative monetary outflow to project i. The Cost Performance Index $\mathrm{CPI}_{i,t} = \mathrm{BCWP}_{i,t}/\mathrm{ACWP}_{i,t}$ measures cost efficiency, while the Schedule Performance Index $\mathrm{SPI}_{i,t} = \mathrm{ES}_{i,t}/\tau_i$ measures schedule efficiency via earned schedule $\mathrm{ES}_{i,t}$—the project-time equivalent of the work accomplished, obtained by inverting the planned S-curve. The Estimate at Completion uses a composite denominator, $\mathrm{EAC}_{i,t} = \mathrm{ACWP}_{i,t} + (\mathrm{BAC}_i - \mathrm{BCWP}_{i,t})/(\mathrm{CPI}_{i,t} \cdot \mathrm{SPI}_{i,t})$, projecting total cost under the assumption that both cost and schedule inefficiency persist. The To-Complete Performance Index $\mathrm{TCPI}_{i,t} = (\mathrm{BAC}_i - \mathrm{BCWP}_{i,t})/(\mathrm{BAC}_i - \mathrm{ACWP}_{i,t})$ measures the cost efficiency required to finish on budget. These four indicators are tracked internally and logged to the audit record each period, but are not directly exposed in the observation vector o_t; instead they are intermediates from which the catchup-allocation and target-milestone features are derived.

Termination. A further distinguishing feature of the proposed environment is that project termination is treated as an *endogenous* consequence of the manager's own funding decisions, rather than an exogenous shock imposed on the simulation. In practice, a contractor can lose a project in two qualitatively different ways: gradually, as sustained underfunding allows schedule slippage to persist past a contractually tolerated threshold and through a grace period intended to allow recovery; or abruptly, once the project's finish date overruns its contractually permitted limit regardless of how close it is to catching up. Capturing only one of these failure modes — as prior single-project formulations typically do — understates the manager's true risk exposure: a project can be lost even while actively improving, simply by running out of contractual time, just as it can survive a severe but short-lived delay provided it recovers before its grace period is exhausted. Termination also carries direct financial consequences that must be settled against the portfolio's cash position, since work already paid for and work already completed rarely coincide exactly at the moment a contract is terminated. By embedding both breach conditions, their associated grace mechanism, and the resulting settlement cash flow directly into the transition dynamics, the environment forces a learned or heuristic policy to treat the threat of termination not as an external constraint to be checked post hoc, but as a consequence of its own allocation choices — precisely the incentive structure a real portfolio manager faces.

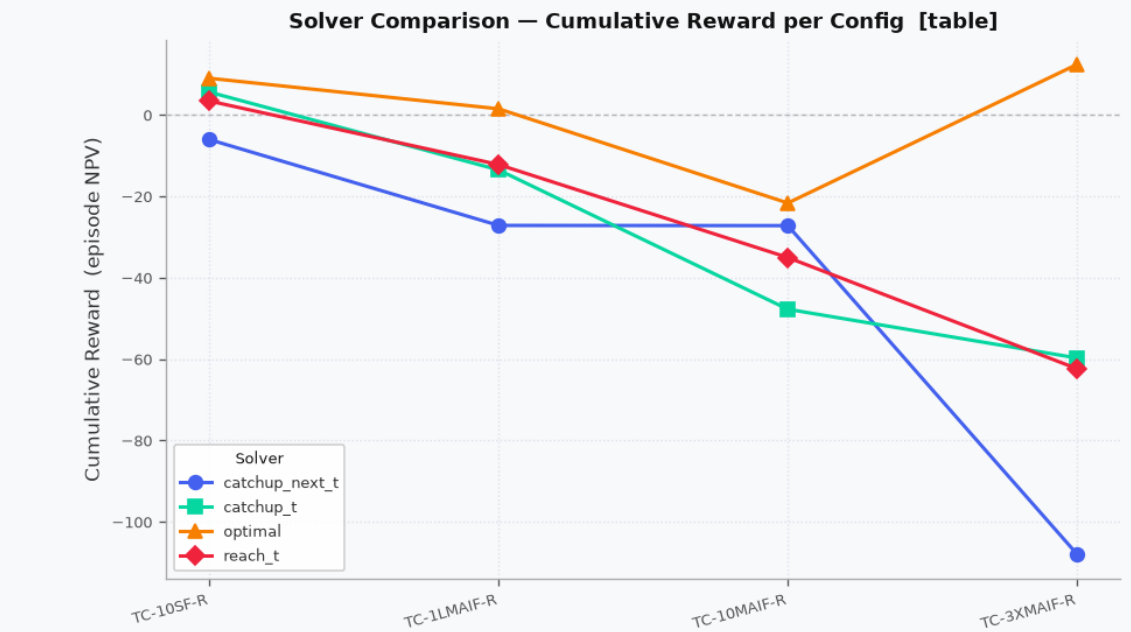
1.4 The Optimality Gap and the Case for Learned Policies

A central empirical question is whether the problem is hard enough to warrant learned policies at all. To answer this, we run three deterministic greedy heuristics against a manually computed full-information optimum across four benchmark portfolio configurations of increasing complexity.

The **Reach-Plan** heuristic (π^{rp}) allocates exactly enough budget each period to bring every active project to its S-curve planned progress, closing the gap entirely. The **Catchup-T** heuristic (π^{cu}) allocates only enough to stay within the contractual progress delay tolerance, minimising expenditure. The **Catchup-Next-T** heuristic ($\pi^{\mathrm{cu}+1}$) is the most risk-averse: it pre-emptively allocates enough this period so that the projected delay next period also remains within tolerance. The **full-information optimum** (π^*) is computed manually with all stochastic realisations known in advance, serving as an upper-bound reference.

Figure 7 reports cumulative episode NPV for each method across four configurations. On the simplest configuration (single project, fixed efficiency) the greedy baselines come within a small margin of the optimum. As portfolio complexity grows — more projects,

staggered starts, interacting cash flows, and milestone sequencing constraints — the gap widens systematically. On the most complex configuration the best greedy baseline trails the optimum by more than 60 NPV units while one baseline degrades to −107.96, a loss-making outcome. This monotone widening confirms that the allocation problem becomes genuinely difficult at scale and that a well-defined improvement target exists for reinforcement learning agents.



	10SF-R	1LMAIF-R	10MAIF-R	3XMAIF-R
catchup_next_t	-6.07	-27.17	-27.22	-107.96
catchup_t	5.60	-13.53	-47.77	-59.75
optimal	9.00	1.49	-21.66	12.40
reach_t	3.40	-12.21	-35.06	-62.38

Figure 2: Cumulative episode NPV (reward) for three greedy heuristics and the full-information optimum across four benchmark configurations of increasing portfolio complexity. The widening gap as complexity grows motivates the use of learned allocation policies.

1.5 Contributions and Paper Organisation

The contributions of this paper are threefold.

C1. Environment. A configurable, Gymnasium-compatible CMDP environment encoding the full FIDIC contract cash-flow cycle: advance payment and progressive recovery, progress-weighted milestone payments, retention withholding and release, treasury-draw interest on negative cash balances that persists until the portfolio returns to surplus, and a two-condition termination gate with a project-specific cure-period counter. All environment parameters are specified through a configuration dictionary whose entries are probability distributions, allowing practitioners in any project-based sector—construction, IT, R&D, or otherwise—to instantiate domain-specific environments by fitting those distributions to historical company data. Gymnasium API compliance is verified across 51 checks in 10

test sections; the full compatibility certificate is reproduced in Appendix B. The environment ships with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag at each episode step, making learned policy behaviour fully interpretable.

C2. MDP Formulation. A formal specification of the portfolio budgeting problem as a CMDP: state space, action space, transition dynamics, reward signal, and the observation vector presented in closed form.

C3. Baseline Benchmark and Optimality Gap. Three deterministic greedy heuristics evaluated against a full-information optimum across four benchmark configurations, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future reinforcement learning agents.

Section 2 reviews the relevant literature across four streams: project management foundations, portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section 3 develops the MDP formulation formally. Section 4 presents the benchmark evaluation: three deterministic greedy heuristics and a full-information optimum across four portfolio configurations, quantifying the optimality gap that motivates learned agents. Training RL agents on the environment and evaluating them against a formal Linear Programming bound are left as the natural next step, addressed in subsequent work.

2 Literature Review

2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review follows the PRISMA 2020 reporting guidelines (Page et al., 2021) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management under uncertainty*, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search. For instance, a high-citation anchor paper on portfolio optimisation exposes its own foundational references through backward traversal, while forward traversal through its citing works surfaces more recent developments; both layers would have been invisible to keyword search alone.

Records were screened in three sequential automated stages consistent with the PRISMA 2020 flow diagram (Appendix A). The screening pipeline, together with its archived corpus and output artefacts, is implemented as a reproducible processing chain whose repository location and concept-level mapping are documented in Appendix C.

Stage 1 — Identification and Machine Readability. Candidate PDFs were subjected to automated bibliographic and readability triage. Each document was assessed for text-extraction yield (minimum readable character density across pages), lexical quality (alphabetic-token ratio and mean word length), and sentence-transformer embeddability. Documents failing any criterion were quarantined in a dedicated rejection archive, with a content hash logged to prevent duplicate submissions from re-entering the pipeline.

Stage 2 — Relevance Filtering. Records passing Stage 1 were ranked by a composite relevance score. A semantic similarity component embedded each paper's abstract, introduction, and conclusion using a sentence-transformer model and computed the maximum cosine similarity against twelve thesis-derived query vectors spanning the target knowledge domains. A keyword-scoring component assigned weighted counts of domain-specific high-value and medium-value terms, with a penalty for domain-irrelevant vocabulary. The two components were combined with equal weight (0.5 each) to yield a combined relevance score $r \in [0, 1]$. Papers were then triaged into three cohorts: *highly relevant* ($r \geq 0.45$), *valuable* ($r \geq 0.30$), and *irrelevant* ($r < 0.30$), with the first two cohorts retained for structured extraction.

Stage 3 — Structured Extraction. Each retained paper in the *highly relevant* and *valuable* cohorts was subjected to structured knowledge extraction. The abstract, introduction, and conclusion of each PDF were parsed and submitted to a large language model with a fixed schema prompt, producing a per-paper record covering: problem framing, portfolio level, decision type, stochastic cash-flow treatment, contract mechanics, termination-settlement provisions, modelling paradigm, calibration parameters (CPI/SPI statistics, cost-overrun distributions, S-curve shape parameters), and thesis-relevance flags. The resulting structured corpus subsequently drove an automated synthesis dashboard and a table generator, producing the gap-analysis matrix, domain-distribution charts, and the literature map reproduced in Table 1.

The complete pipeline — processing scripts, the structured record corpus, dashboard outputs, and synthesis tables — is archived within the project repository and constitutes a reproducible, machine-auditable literature review corpus; its directory-level organisation is detailed in Appendix C. The PRISMA flow diagram quantifying records at each stage is provided in Appendix A.

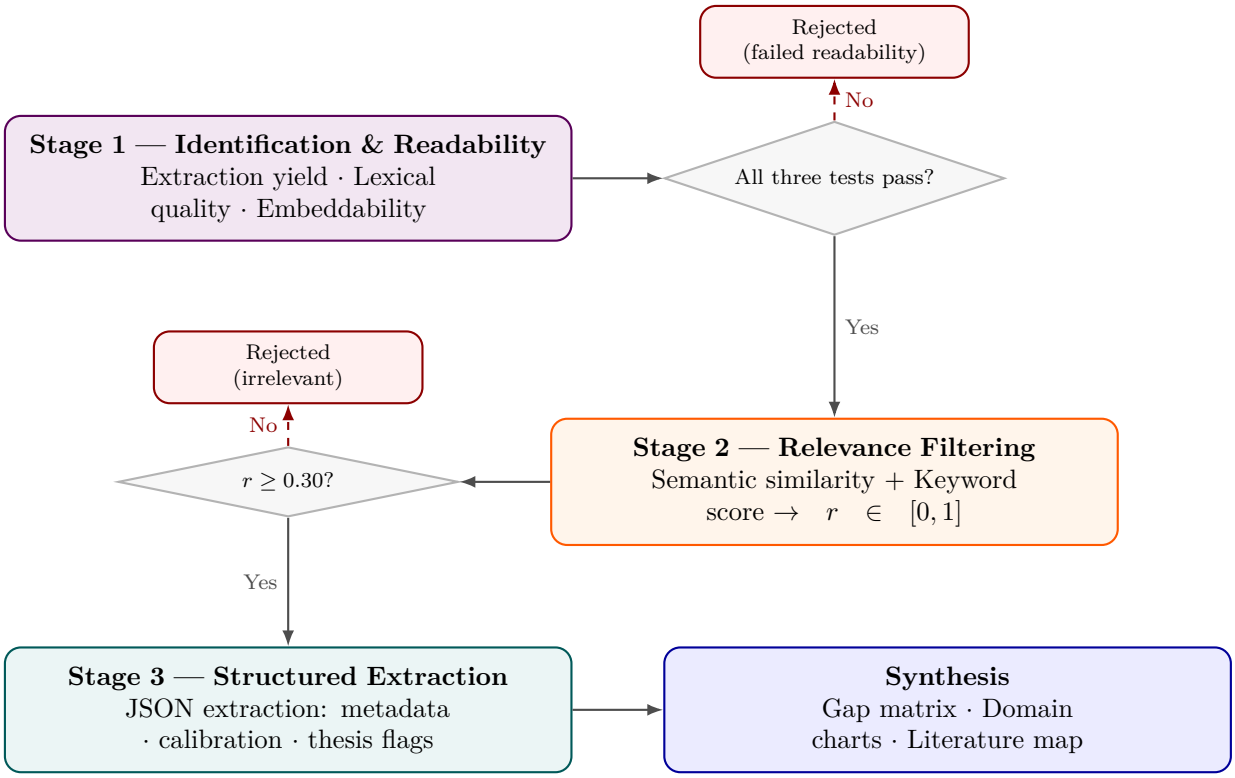


Figure 3: Three-stage PRISMA-aligned screening pipeline, arranged as a serpentine flow. Boxes denote automated processing stages; diamonds denote pass/fail gates; dashed arrows indicate rejection routing. Implementation details and repository locations are provided in Appendix C.

2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (Fleming and Koppelman, 2010; Project Management Institute, 2021). EVM introduced the Cost Performance Index (CPI) and Schedule Performance Index (SPI) as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. The present model adopts the Earned Schedule correction to SPI (Lipke, 2003), replacing the classical ratio BCWP/BCWS with SPI(t) = ES/AT, which continues to degrade past the planned finish date rather than freezing when BCWS reaches BAC — a known limitation of classical SPI in post-planned-finish recovery periods (Lipke, 2003; **?**). The To-Complete Performance Index (TCPI) is also included as a forward-looking complement to CPI, measuring the cost efficiency required on remaining work to finish within budget. Values of CPI < 1 signal cost overrun; SPI < 1 signals schedule lag. Stochastic extensions of EVM (Acebes et al., 2014; Batselier and Vanhoucke, 2017) and Monte Carlo simulation frameworks (Kwak and Ingall, 2007) have modelled performance uncertainty at the project level, establishing that CPI and SPI follow empirically characterisable distributions over project life. Empirical evidence on contractor performance variability from EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017) provides the calibration basis for the stochastic efficiency parameter $\eta_{i,t}$ that parameterises the stochastic program at Level 2 and the CMDP environment at Level 3.

S-curve analysis, formalised by Kenley and Wilson (2009) for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularized incomplete Beta function provides a flexible single-family parameterisation of this profile (Barraza et al., 2004), and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by Merrow (2011) through the Independent Project Analysis database, and by Flyvbjerg et al. (2003); Flyvbjerg (2017) through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions in the present model and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (Christensen and Heise, 1993).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (Fédération Internationale des Ingénieurs-Conseils, 2017). The Estimate at Completion $\mathrm{EAC}_{i,t} = \mathrm{ACWP}_{i,t} + (\mathrm{BAC}_i - \mathrm{BCWP}_{i,t})/\mathrm{CPI}_{i,t}$ extends CPI into a forward-looking cost trajectory signal; Christensen and Heise (1993) established across 155 contracts that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (Lipke, 2003). The present model adopts EAC as the second termination condition signal, directly parallel to the forecast finish deviation used in Condition 1. Under FIDIC Sub-Clause 14.2, the advance payment is an interest-free mobilisation loan recovered through percentage deductions from interim payment certificates: if not otherwise specified in the Appendix to Tender, deductions commence once cumulative certified interim payments exceed 10% of the Accepted Contract Amount and are applied at a rate of 25% of each subsequent payment certificate until the advance is fully recovered (Fédération Internationale des Ingénieurs-Conseils, 2017). The present model parameterises this mechanism with a project-specific commencement threshold ψ_i and deduction rate δ_i^{rec}, accommodating the full range of particular conditions encountered in practice. These mechanisms translate directly into the cash inflow and outflow dynamics formulated in Section **??**.

2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation's finite resources across a competing set of candidate projects to maximise strategic value (Archer and Ghasemzadeh, 1999; Cooper et al., 2001). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean–variance frameworks adapted from financial portfolio theory (Markowitz, 1952) have been applied to project baskets to trade off expected NPV against variance (Eilat et al., 2006). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (Martinsuo and Lehtonen, 2007; Killen et al., 2008). Metaheuristic approaches — genetic algorithms, particle swarm optimisation — address large combinatorial instances (Zhang et al., 2020). Resource-constrained portfolio selection under capital rationing has been studied by Kavadias and Loch (2003), whose

no-borrowing constraint is adopted directly in the present formulation.

The present work *explicitly adopts the output of portfolio selection as an exogenous input.* The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (Kenley and Wilson, 2009; Park and Peña-Mora, 2004), EVM-based forecasting of individual project cost-at-completion (Kim and Ballard, 2002; Acebes et al., 2014), and multi-project resource levelling (Lova et al., 2000; Van Peteghem and Vanhoucke, 2010). However, these contributions treat budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in the present model arises from a single source: the budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period (Acebes et al., 2014; Batselier and Vanhoucke, 2017). Uncertain productivity determines whether and when each milestone threshold is crossed, which in turn determines the timing of the contractually fixed payment $R_{i,j}^{\text{net}}$. Payment delays between certification and receipt are not modelled as an independent uncertainty source. This relaxation is defensible: the performance-driven channel is the structurally primary source of cash-flow timing uncertainty in milestone-payment contracts, and isolating it yields a cleaner identification of the allocation problem without loss of the essential uncertainty framing. Other sources of cash-flow uncertainty — scope changes, external market shocks (Doloi, 2012) — are treated as exogenous and outside the scope of the present formulation. Stochastic programming (Birge and Louveaux, 2011) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (Lova et al., 2000). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases. The single uncertainty source $\eta_{i,t}$ now branches the scenario tree at every period: if $\eta_{i,t}$ takes K discrete values at each of H periods for each of n projects, the tree produces K^{nH} leaf nodes. The effective planning horizon is itself portfolio-dependent—determined by the latest project finish—compounding the explosion, since H grows with the most delayed project in the committed set. Both effects are formalised in Section **??**.

The gap in the literature is precise: *no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow trajectories are uncertain through stochastic budget-to-progress efficiency, subject to contractual milestone certification windows that bound the speed at which progress converts to cash inflows.* The present work addresses this gap. The objective is pure discounted net cash flow; no deviation penalty enters the objective function. This is a deliberate design

choice: the consequences of sustained deviation — termination and loss of all future milestone payments and retention — are sufficient to make deviation undesirable without a separate penalty term, and including one would introduce an arbitrary scalarisation weight with no natural calibration from contract data (Ng et al., 1999). Symmetrically, the discount factor γ encodes not only time preference but the opportunity cost of tied-up capital in a competitive environment where future projects are foregone while current ones remain incomplete (Damodaran, 2012). This interpretation gives γ a concrete calibration anchor beyond abstract preference — it can be estimated from the firm's observed pipeline redeployment rates or from prevailing cost-of-capital benchmarks. The termination settlement further grounds the model in real asset values: the settlement payment $R_i^{\text{term}} = P_{i,t}^{\text{actual}} \cdot CP_i - (A_i + \sum_{j \in \mathcal{M}_i^{\text{cert}}} R_{i,j}^{\text{net}})$ computes the difference between the contractor's entitlement for work delivered and cumulative inflows already received, which can be positive or negative depending on progress at termination relative to certified payments (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), rather than an arbitrarily weighted deviation penalty. A consequence is that the learned policy may produce allocation trajectories that concentrate expenditure around milestone certification thresholds, mirroring documented contractor financial strategy in milestone-payment contracts (Kenley and Wilson, 2009; Kaka and Price, 1993; Arditi and Chotibhongs, 2005).

2.5 Contractor Underperformance, Cure Periods, and Termination

When a contractor fails to meet schedule or cost performance thresholds, international EPC contracts prescribe a structured escalation sequence before termination may be exercised. Under FIDIC Sub-Clause 15.1, the client must issue a formal notice of default and allow a cure period — typically 28 days — during which the contractor is obliged to demonstrate measurable recovery before the client may proceed to termination (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008). This cure period is not a discretionary grace; it is a contractual obligation whose omission constitutes wrongful termination. Empirical studies of international EPC disputes confirm that failure to observe the cure period is among the most frequently litigated grounds in contractor termination claims (Ndekugri and Rycroft, 2008; Ennis and Burr, 2021).

The behavioural project management literature has studied the conditions under which organisations fail to terminate underperforming projects at the economically rational point. Escalation of commitment (Staw and Ross, 1987) — the tendency to continue investing in failing projects beyond what expected value justifies — has been documented across construction, IT, and infrastructure programmes. Real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001) provides the normative counterpart: termination is the exercise of an abandonment option whose value exceeds continuation NPV once performance has deteriorated past a recoverable threshold. Together these streams establish that the decision to persist with or terminate an underperforming project is neither purely contractual nor purely financial — it involves performance signals, contractual obligations, and timing all at once.

The present model integrates all three dimensions into a single environment mechanic governed by two conditions. A *progress delay breach* fires whenever the lag between planned and actual progress exceeds the contractual tolerance $\delta_i^{\max}$, i.e. $\delta_{i,t} = PV_i(\tau_i) - EV_{i,t} > \delta_i^{\max}$, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (Fédération Internationale des Ingénieurs-Conseils, 2017). A *hard deadline breach* fires unconditionally at episode period $t \geq f_i + \Delta_i^{\max}$, where $\Delta_i^{\max}$ is the contractually permitted finish overrun, and triggers immediate termination regardless of the cure-period counter state. When a progress delay breach is active,

the cure-period counter $\theta_{i,t}^{\text{rem}}$ decrements by one; it resets to θ_i only when no breach is present. Termination executes when the counter reaches zero. This mechanic is grounded simultaneously in FIDIC contract law (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008), the real options framing of the abandonment threshold (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), the EVM literature's validation of EAC as a stable forward cost signal (Christensen, 1993; Lipke, 2003), and the escalation-of-commitment literature's evidence that observable performance signals must be sustained — not transient — before termination is justified (Staw and Ross, 1987). The counter $\theta_{i,t}^{\text{rem}}$ is part of the observable portfolio state, giving the RL agent an anticipatory signal that reflects contractual reality without granting it discretionary control over the termination outcome. From the portfolio manager's perspective, this signal serves a dual function: it flags projects approaching contractual failure while also marking the point at which continued allocation yields diminishing returns relative to the settlement value of work already delivered. A rational manager reading a counter approaching zero in a high-opportunity-cost environment would recognise that the real value of future milestone payments is eroding while the settlement value of work in place is fixed in nominal terms — creating a natural reallocation incentive that the model captures through the interaction of γ and $S_i = EV_{i,t} \cdot P_i - \text{CF}_i^+$ without requiring explicit inflation modelling (Dixit and Pindyck, 1994; Fédération Internationale des Ingénieurs-Conseils, 2017).

2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (Sutton and Barto, 2018). The MDP formalisation — state, action, transition, reward — maps naturally onto the portfolio budget allocation problem, and the constrained MDP extension (Altman, 1999) accommodates the hard budget feasibility constraints that the portfolio manager must satisfy at every period.

RL has been applied to project scheduling (Peng and Huang, 2021; Liu et al., 2023), resource-constrained project scheduling problems (RCPSP) (Tassel et al., 2023), and supply chain inventory management (Giannoccaro and Pontrandolfo, 2002). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, the problem is non-stationary over the decision horizon, and the horizon length is itself uncertain at the outset of planning. These are precisely the characteristics of the portfolio budget allocation problem at Levels 2 and 3.

Proximal Policy Optimisation (Schulman et al., 2017) is adopted as the solution algorithm for its stability in continuous action spaces, its sample efficiency relative to earlier policy-gradient methods, and its established use in operations management domains. Domain randomisation (Tobin et al., 2017) — randomising environment parameters during training to improve out-of-sample policy robustness — is adopted to ensure the learned policy generalises across the range of project types, contract structures, and market conditions represented in the calibration data. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data.

Direct application of RL to portfolio-level budget allocation remains absent from the literature. The present work provides the first formulation of this problem as a CMDP and the first empirical demonstration that a PPO agent can learn non-trivial allocation policies in a calibrated synthetic environment.

2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework, as generated by the automated synthesis pipeline described in Section 2.1.1. Three structural gaps emerge from this mapping.

Gap 1 — Selection versus operational governance. The transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint. No existing work closes both simultaneously.

Gap 2 — Project-level versus portfolio-level uncertainty propagation. Existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not propagate uncertainty upward into a portfolio-level decision architecture governed by a shared budget constraint. The feedback loop between a project's stochastic efficiency trajectory and its portfolio-level funding entitlement remains unmodelled.

Gap 3 — Absence of endogenous termination mechanics. The joint modelling of a progress-delay breach condition tracked against a contractual tolerance, a decrementing cure-period counter governing grace-period exhaustion, an independent hard-deadline breach condition that triggers immediate termination regardless of the cure-period state, and a termination settlement cash flow reconciling earned value against cumulative payments received, has not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 4 situates this progression as three sequential steps.

Step 1 — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires.

Step 2 — tractable problem formulation — is the gap this paper closes. The contribution is a verified, configurable CMDP environment that encodes the full FIDIC cash-flow cycle and a two-condition termination mechanic, released as a Gymnasium-compatible package with a complete SQLite audit ledger. Baseline greedy heuristics are benchmarked against a full-information optimum, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future learned agents.

Step 3 — validated deployment within incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the trained agents that Step 2 now makes possible.

The present work addresses all three gaps and completes Step 2 of this progression. The Gymnasium-compatible CMDP environment — with parameterisable contract mechanics, stochastic budget-to-progress efficiency as the single uncertainty source, and an endogenous termination mechanic — constitutes the primary methodological contribution. Baseline heuristics are evaluated against a full-information optimum in Section 4, providing a quantified optimality gap that grows monotonically with portfolio complexity and defines the improvement target for future RL agents.

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ○ = partially addressed; – = not addressed.

Work	Portfolio level	Decision type	Sequential decisions	Stochastic cash flows	Contract mechanics	Termination settlement
Yamakawa et al. (2019)	✓	S	–	○	–	–
Kosztyán et al. (2022)	✓	S	–	○	–	–
Touran (2006)	✓	Sc/B	–	○	–	–
Touran (2010)	✓	Sc/B	–	○	–	–
Ghapanchi et al. (2012)	✓	S	–	○	–	–
Pourahmadi et al. (2015)	✓	S	–	○	–	–
El-Sabek and McCabe (2018)	–	B	–	–	✓	–
Beseiso and Kumar (2021)	✓	S	–	○	–	–
Karimi et al. (2021)	✓	S	–	○	–	–
Bahroun et al. (2023)	–	Sc	✓	–	–	–
Ranjbar et al. (2022)	✓	S/Sc	–	–	○	–
Szwarcfiter et al. (2023)	✓	Sc	✓	○	–	–
Asghari et al. (2023)	–	–	✓	○	–	–
Mazhari and Tajik (2024)	✓	S/B	✓	–	–	–
Jiang et al. (2024)	✓	–	✓	○	–	–
Nabot et al. (2025)	✓	Sc	✓	–	✓	✓
Darvish and Sepehri (2025)	✓	B	✓	○	○	✓
Gholami et al. (2025)	✓	Sc	✓	○	–	✓
Kehinde (2025)	✓	B	✓	✓	–	–
Ajibade (2025)	✓	Sc	✓	○	–	–
Huddar and Joshi (2025)	✓	B	✓	○	–	–
Akbari and Shirazi (2026)	✓	B	✓	○	✓	✓
Ebnerasoul et al. (2023)	✓	S	–	○	–	–
Noroozi et al. (2016)	✓	B	–	○	–	–
Jiang et al. (2023)	–	B	✓	–	–	–
Present work	✓	**B**	✓	✓	✓	✓

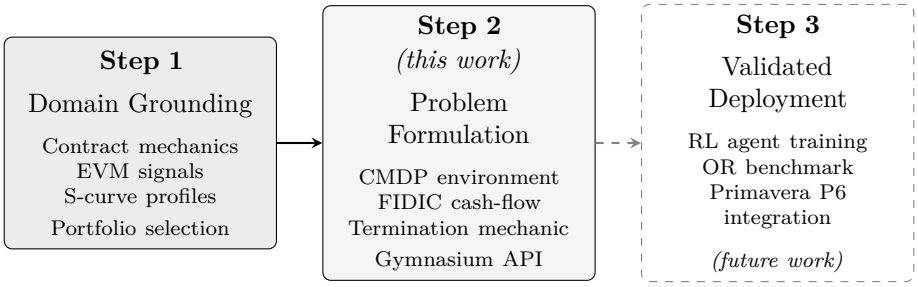


Figure 4: Maturity progression toward a deployable portfolio-level decision-support system. Step 2, completed by the present work, enables Step 3. Dashed border and arrow denote future work.

3 Problem Formulation

This section develops the complete mathematical formulation of the portfolio budget allocation problem and its specification as a Gymnasium-compatible reinforcement learning environment. The section is structured as follows. Section 3.1 states the problem scope and structural assumptions. Section 3.2 defines the environment primitives: project dynamics, FIDIC contract cash-flow mechanics, EVM performance signals, the termination mechanic, and portfolio balance. Section 3.3 consolidates all parameters. Section 3.4 establishes the deterministic full-foresight upper bound and the information-theoretic argument for why the stochastic problem cannot be solved by enumeration. Section 3.5 formulates the problem as a Markov Decision Process (MDP), defining the state space, action space, transition kernel, and reward function that constitute the environment's mathematical contract.

3.1 Problem Statement and Scope

A portfolio of n contracted projects is legally fixed. commitments are signed, start dates are scheduled, and contracts are parameterised consistently with FIDIC Red, Yellow, and Silver Book forms (Fédération Internationale des Ingénieurs-Conseils, 2017). The decision-maker — a portfolio manager, board, or programme director — exercises a single recurring lever at each discrete planning period $t \in \mathcal{T} = \{0, 1, \ldots, H-1\}$: the budget $x_{i,t} \geq 0$ directed to each active project i. The planning horizon H is determined endogenously as $H = \max_i(f_i + \Delta_i^{\max})$, where f_i is the planned finish of project i and $\Delta_i^{\max}$ is the maximum permitted overrun; period $t = 0$ aligns with the earliest project start. All outcomes of interest — milestone achievement, cash inflows, schedule adherence, and project termination — flow through this allocation decision. Portfolio selection is treated as an exogenous upstream input (Archer and Ghasemzadeh, 1999).

Four structural assumptions underlie the formulation.

(i) *Independent bilateral contracts.* Projects interact only through the shared cash balance; no resource or workforce sharing occurs, consistent with standard EPC contracting practice.

(ii) *Budget as the sole control.* The contractor's internal resource decisions are absorbed into a stochastic efficiency parameter $\eta_{i,t}$, it determines the ratio of the budget to be verified as progress; the portfolio manager observes how much physical progress a budget allocation yields and then allocates budget to each project. $\eta_{i,t}$ is the sole stochastic primitive of the model. it is unobservable at the time allocation

$x_{i,t}$ is committed, and its realisation determines whether and when each milestone threshold is crossed. All other parameters are either known at episode initialisation or deterministically derived from prior realisations.

(iii) *Discrete monthly periods.* $\mathcal{T} = \{0, \ldots, H-1\}$ reflects the standard monthly governance cycle of portfolio management. The scalar $\kappa = B_0 / \sum_i \mathrm{BAC}_i$ parameterises the liquidity regime across configurations: values below 1 force the agent to compete over a scarce pool, while values above 1 provide budget headroom beyond aggregate project demand. In this work κ is set to a relaxed value.

(iv) *FIDIC project contract management.* Milestone payments, advance payment and recovery, retention, and termination settlement are all configurable parameters, accommodating the full range of particular conditions encountered in practice.

3.2 Environment Definition

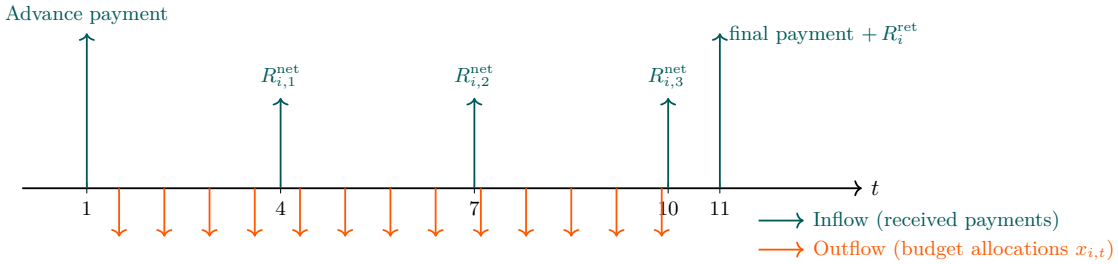


Figure 5: Cash-flow timeline for project i under FIDIC contract mechanics with uniform allocation. The ratio of budget allocated to physical progress achieved is subject to stochastic efficiency $\eta_{i,t}$, the sole source of uncertainty in the model.

3.2.1 Sets, Indices, and the Decision Variable

$\mathcal{P}$	Projects $\{0, \ldots, n-1\}$, indexed by i; all committed at horizon start.
$\mathcal{T}$	Planning periods $\{0, \ldots, H-1\}$, indexed by t (calendar months), where $H = \max_{i \in \mathcal{P}}(f_i + \Delta_i^{\max})$ is derived endogenously from the committed portfolio; period $t = 0$ aligns with the earliest project start.
$\mathcal{M}_i$	Ordered milestones $\{1, \ldots, M_i\}$ for project i, indexed by j, with progress thresholds $\theta_{i,1} < \cdots < \theta_{i,M_i} = 1$. advance payment is recorded as M_0 at project start s_i with progress threshold zero.
$\mathcal{P}_t^{\text{active}}$	Projects with `active` status at period t; endogenous to the efficiency trajectory and termination events. Projects get activated by their start and stay active as long as they stay healthy. only active projects accept allocation action and provide milestone payment.
$x_{i,t} \geq 0$	**Decision variable**: budget allocated to project i at period t. The period-t action vector is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$.
$B_t \geq 0$	Portfolio cash balance at the start of period t; B_0 is given.

3.2.2 Progress and Expenditure Dynamics

Budget $x_{i,t}$ converts to physical progress through the efficiency scalar $\eta_{i,t}$, which absorbs all sources of conversion uncertainty (contractor productivity, supply chain, weather):

$$P_{i,t}^{\text{actual}} = \min\left(1,\ P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}\right), \qquad \text{ACWP}_{i,t} = \text{ACWP}_{i,t-1} + x_{i,t}, \tag{1}$$

with $P_{i,0}^{\text{actual}} = \text{ACWP}_{i,0} = 0$. Here BAC_i is the Budget at Completion — the total planned contractor expenditure for project i, which serves as the EVM cost basis and the normalisation denominator for all progress calculations. The efficiency $\eta_{i,t}$ is realised from a project-specific distribution only after allocation $x_{i,t}$ is committed; it is the sole stochastic primitive. A global floor of $\eta_{i,t} \geq 0.01$ prevents degenerate zero-progress episodes.

The planned cumulative progress follows a normalised S-curve (Kenley and Wilson, 2009; Barraza et al., 2004) parameterised by Beta shape parameters (a_i, b_i). Defining the normalised project-time fraction $\tau_{i,t} = (t - s_i)/D_i^{\text{plan}}$ clipped to $[0, 1]$, the planned progress at period t is:

$$P_{i,t}^{\text{plan}} = I_{\tau_{i,t}}(a_i, b_i), \tag{2}$$

where $I_\tau(\cdot, \cdot)$ is the regularised incomplete Beta function and $D_i^{\text{plan}} = f_i - s_i$ is the planned duration. This profile is a fixed parameter derived at commitment; it is never a decision variable.

3.2.3 Contractual Cash-Flow Mechanics

Payment inflows follow FIDIC mechanics (Fédération Internationale des Ingénieurs-Conseils, 2017). At project initiation ($t = s_i$), the portfolio manager receives an advance payment:

$$A_i = \alpha_i \cdot CP_i, \tag{3}$$

where α_i is the advance ratio and $CP_i = \text{BAC}_i(1 + \pi_i)$ is the contract price with profit margin π_i. The contract price is the sole payment accumulation baseline: all milestone revenues, advance payment, retention, and final settlement sum to CP_i. All EVM performance signals and progress calculations are normalised against BAC_i.

Milestone $j \geq 1$ is certified at the first period t at which both $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ and $t \geq e_{i,j}$, where $e_{i,j} = s_i + \tau_{i,j}^{\text{thresh}}$ is the contractual earliest certification period and $\tau_{i,j}^{\text{thresh}}$ is the project-time threshold (Fédération Internationale des Ingénieurs-Conseils, 2017). Certification is monotone: once certified, a milestone remains certified. The net payment received upon certification of milestone j, after retention withholding at rate ρ_i and advance recovery deduction at rate δ_i^{rec} once cumulative certified payments exceed threshold ψ_i, is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i - \rho_i \cdot \phi_{i,j} \cdot CP_i - \delta_i^{\text{rec}} \cdot \phi_{i,j} \cdot CP_i \cdot \mathbf{1}\left\{\textstyle\sum_{k \leq j} \phi_{i,k} \geq \psi_i\right\} + \rho_i \cdot CP_i \cdot \mathbf{1}\{j = M_i\}, \tag{4}$$

where $\phi_{i,j}$ is the contractual weight of milestone j with $\sum_j \phi_{i,j} = 1$. The final term releases accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at project completion. The payment identity $A_i + \sum_j R_{i,j}^{\text{net}} = CP_i$ holds exactly. Milestone payments are received in the same period they are certified. The aggregate period inflow is:

$$V_t = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\tau_{i,j}^{\text{cert}} = t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \tag{5}$$

where $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t and R_i^{term} is defined in Section 3.2.5.

3.2.4 Performance Monitoring: EVM Signals

The Earned Value Management framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary used to monitor contractor performance and to drive the termination mechanic. The Earned Value at period t is $\mathrm{EV}_{i,t} = P_{i,t}^{\mathrm{actual}} \cdot \mathrm{BAC}_i$. The project-time elapsed since project start is denoted $\tau_i^{\mathrm{proj}}(t) = t - s_i$. Four derived indices are computed each period.

The Schedule Performance Index uses the Earned Schedule formulation (Lipke, 2003). The Earned Schedule $\mathrm{ES}_{i,t}$ is the value $\tau^* \in [0, D_i^{\mathrm{plan}}]$ satisfying $I_{\tau^*/D_i^{\mathrm{plan}}}(a_i, b_i) = P_{i,t}^{\mathrm{actual}}$, computed by binary search over the Beta CDF:

$$\mathrm{SPI}_{i,t} = \frac{\mathrm{ES}_{i,t}}{\tau_i^{\mathrm{proj}}(t)}, \tag{6}$$

$$\mathrm{CPI}_{i,t} = \frac{\mathrm{EV}_{i,t}}{\mathrm{ACWP}_{i,t}}, \tag{7}$$

$$\mathrm{EAC}_{i,t} = \mathrm{ACWP}_{i,t} + \frac{\mathrm{BAC}_i - \mathrm{EV}_{i,t}}{\mathrm{CPI}_{i,t} \cdot \mathrm{SPI}_{i,t}}, \tag{8}$$

$$\mathrm{TCPI}_{i,t} = \frac{\mathrm{BAC}_i - \mathrm{EV}_{i,t}}{\mathrm{BAC}_i - \mathrm{ACWP}_{i,t}}. \tag{9}$$

Unlike the classical SPI = EV/PV, which freezes once the planned value reaches BAC_i at planned finish, the Earned Schedule formulation (6) degrades continuously as real time passes without progress, correctly distinguishing a project one period late from one three periods late (Lipke, 2003). This formulation is adopted throughout. The composite denominator $\mathrm{CPI}_{i,t} \cdot \mathrm{SPI}_{i,t}$ in (8) incorporates both cost and schedule underperformance into the cost-at-completion forecast. $\mathrm{TCPI}_{i,t}$ measures the cost efficiency that must be sustained on remaining work to finish within BAC_i; values above 1.10 are widely considered unrecoverable (Fleming and Koppelman, 2010; Christensen, 1993). All four indices are observable state components. Denominators that would produce undefined results default to one (no overrun implied), consistent with the project state before any expenditure has been committed.

3.2.5 Termination Mechanic

Termination is an endogenous environment mechanic, not an agent action. Two conditions are evaluated for every active project at each period.

Condition 1 Progress delay breach. The progress lag at period t is:

$$\delta_{i,t} = P_{i,t}^{\mathrm{plan}} - P_{i,t}^{\mathrm{actual}}. \tag{10}$$

Condition 1 activates when this lag exceeds the contractual cap:

$$\delta_{i,t} > \delta_i^{\max}, \tag{11}$$

where $\delta_i^{\max} \in [0, 1]$ is the project-specific tolerance fixed at commitment, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (Fédération Internationale des Ingénieurs-Conseils, 2017).

Condition 2 Hard deadline breach. A hard deadline fires unconditionally at:

$$t \geq f_i + \Delta_i^{\max}, \tag{12}$$

where $\Delta_i^{\max} \in \mathbb{Z}^+$ is the maximum permitted finish overrun in periods. This triggers immediate termination regardless of the cure-period counter state.

Cure-period counter and termination execution. When Condition 1 is active the environment decrements a project-specific cure-period counter, modelling the FIDIC Sub-Clause 15.1 obligation to allow a recovery interval before termination may be executed (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008):

$$\tau_{i,t}^{\text{rem}} = \begin{cases} \max\left(0,\ \tau_{i,t-1}^{\text{rem}} - 1\right) & \text{if (11) holds,} \\ \tau_i^{\text{tol}} & \text{if (11) lapses.} \end{cases} \tag{13}$$

The counter initialises at $\tau_{i,s_i}^{\text{rem}} = \tau_i^{\text{tol}}$, where $\tau_i^{\text{tol}} \in \mathbb{Z}^+$ is fixed at commitment. Soft termination executes when the counter reaches zero: $t_i^{\text{term}} = \min\{t : \tau_{i,t}^{\text{rem}} = 0\}$. Hard termination (12) overrides the counter and executes immediately. Condition 2 is evaluated in the early phase of each period (before allocation); Condition 1 and counter updates apply in the late phase (after allocation and progress realisation). The counter is part of the observable state; the agent influences it indirectly through allocations that drive $P_{i,t}^{\text{actual}}$ relative to $P_{i,t}^{\text{plan}}$.

Upon termination at t_i^{term}, the net settlement is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i - \text{cumInflow}_{i,t_i^{\text{term}}}, \tag{14}$$

where $\text{cumInflow}_{i,t}$ is the cumulative inflow received for project i through period t (advance payment plus all net milestone payments certified up to and including t). The first term is the contractor's entitlement for work physically delivered; the second is what has already been collected. Positive R_i^{term} arises when delivered work exceeds cumulative receipts; negative R_i^{term} constitutes a net contractor obligation (large advance not yet recovered by certified milestones). In either case the settlement is fully determined by contract parameters and the progress level at termination.

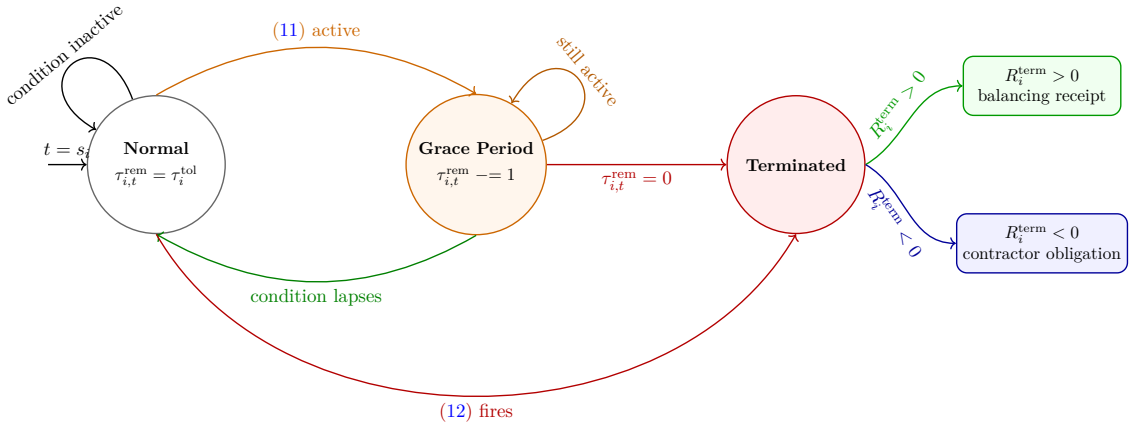


Figure 6: Finite-state representation of the termination mechanic for project i.

3.2.6 Portfolio Cash Balance

The portfolio cash balance evolves as:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t. \tag{15}$$

The shared balance B_t is the sole coupling across otherwise independent project dynamics. The budget tightness ratio $\kappa = B_0 / \sum_i \text{BAC}_i$ parameterises the liquidity regime: $\kappa \gg 1$ (relaxed, liquidity non-binding), $\kappa \approx 1$ (tight), $\kappa \ll 1$ (scarce). When $B_t \leq 0$ and no advance payments are pending, the episode truncates immediately as no further progress is achievable.

3.3 Consolidated Parameter Reference

Table 2 lists all parameters fixed at commitment, grouped by function.

3.4 The Deterministic Upper Bound and the Case for MDP

The problem as stated — choosing $\mathbf{x}_t$ under unknown $\eta_{i,t}$ — is a multi-stage stochastic optimisation problem. Establishing a reference against which any policy can be measured requires a tractable upper bound.

3.4.1 Full-Foresight Upper Bound

Consider a hypothetical oracle that knows the realised efficiency sequence $(\eta_{i,t})_{i,t}$ before making any allocation decision. Under deterministic efficiency, $P_{i,t}^{\text{actual}}$ is a deterministic function of $\mathbf{x}_t$, and therefore all milestone certification times and termination periods are computable in advance from the allocation sequence. The oracle solves the discounted net cash-flow maximisation:

$$Z^* = \max_{\{x_{i,t} \geq 0\}} \sum_{t=0}^{H-1} \gamma^t \left(V_t^{\text{det}} - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right), \tag{16}$$

subject to $\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t$ for all t, and the progress recursion (1) with $\eta_{i,t}$ replaced by the known realised values. All certification timing and termination events in V_t^{det} are deterministic functions of the allocation sequence under known efficiency.

Proposition 1 (Full-foresight upper bound)**.** *Let Z^* denote the value of (16) on a realised efficiency trajectory. Let Z^π denote the objective value achieved by any causal policy π — one that commits $\mathbf{x}_t$ using only information available at time t — on the same trajectory. Then:*

$$\sup_{\pi \in \Pi^{\text{causal}}} \mathbb{E}[Z^\pi] \leq Z^*, \tag{17}$$

where the expectation is over realised efficiency trajectories.

Proof. Fix any realised trajectory $(\eta_{i,t})$. Any allocation sequence produced by a causal policy on this trajectory is also feasible in (16) solved on the same realised values — the feasibility constraints are identical, and the oracle has strictly more information. Therefore $Z^\pi \leq Z^*$ pointwise, and taking expectations preserves the inequality. □

Parameter	Symbol	Domain	Definition
Schedule			
Planned start & finish	$s_i,\ f_i$	$\mathbb{Z}_0^+$	Anchor of execution window and S-curve.
Planned duration	D_i^{plan}	$\mathbb{Z}^+$	$f_i - s_i$; normalises project-time for S-curve.
S-curve shape	$a_i,\ b_i$	$\mathbb{R}^+$	Beta CDF parameters governing planned expenditure profile (2).
Milestone progress threshold	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j; $\theta_{i,M_i} = 1$.
Milestone timestep threshold	$\tau_{i,j}^{\text{thresh}}$	$\mathbb{Z}_0^+$	Project-time period at which milestone j becomes eligible for certification; enforces FIDIC Sub-Clause 14.3 payment interval constraints (Fédération Internationale des Ingénieurs-Conseils, 2017).
Progress delay cap	$\delta_i^{\max}$	$[0, 1]$	Maximum admissible progress lag before Condition 1 (11) activates.
Finish delay cap	$\Delta_i^{\max}$	$\mathbb{Z}^+$	Maximum permitted finish overrun in periods before hard deadline (12) fires.
Termination tolerance	τ_i^{tol}	$\mathbb{Z}^+$	Cure-period length; models FIDIC Sub-Clause 15.1 (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008).
Contract			
Contract price	CP_i	$\mathbb{R}^+$	Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$.
Budget at completion	BAC_i	$\mathbb{R}^+$	Total planned contractor expenditure; EVM cost basis and progress normalisation denominator.
Profit margin	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i.
Advance payment ratio	α_i	$[0, 1]$	$A_i = \alpha_i \cdot CP_i$ received at $t = s_i$.
Recovery deduction rate	δ_i^{rec}	$(0, 1]$	Fraction of each interim payment deducted toward advance recovery once threshold ψ_i is exceeded.
Recovery threshold	ψ_i	$[0, 1]$	Cumulative certified payments as fraction of CP_i at which advance recovery begins.
Retention ratio	ρ_i	$[0, 1]$	Fraction withheld per milestone; released as $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at completion.
Milestone weight	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i at milestone j; $\sum_j \phi_{i,j} = 1$.
Portfolio			
Discount factor	γ	$(0, 1]$	Per-period discount factor; encodes time preference and opportunity cost of capital (Damodaran, 2012); shared across $\mathcal{P}$.
Initial cash balance	B_0	$\mathbb{R}^+$	Portfolio cash balance at horizon start.
Planning horizon	H	$\mathbb{Z}^+$	Derived as $\max_i(f_i + \Delta_i^{\max})$; not a fixed input.
Budget tightness	κ	$\mathbb{R}^+$	$B_0 / \sum_i \text{BAC}_i$; defines the budget regime.
Stochastic Primitive			
Budget-to-progress efficiency	$\eta_{i,t}$	$\mathbb{R}^+$	Sole stochastic primitive. Realised after $x_{i,t}$ is committed; drawn from a project-specific distribution (configurable: fixed, uniform, Beta, triangular, custom ordinal sequence). Floor:

The gap $Z^* - \mathbb{E}[Z^\pi]$ quantifies the combined cost of uncertainty and decision-making quality for policy π. This gap is the primary performance metric of Section 5. In the present work, Z^* is computed manually by solving individual benchmark configurations by hand, with full knowledge of the efficiency sequence realised during each play session. These hand-solved solutions constitute the theoretical upper bound against which all benchmark policies are compared.

3.4.2 Intractability of Exact Stochastic Solution

The exact stochastic counterpart of (16) — maximising the expectation over $\eta_{i,t}$ realisations subject to a causal information structure — is a multi-stage stochastic program (Birge and Louveaux, 2011). The canonical solution method is scenario-tree enumeration. Discretising $\eta_{i,t}$ to K values per project per period, the tree has K^{nH} leaf nodes. For a representative portfolio with $n = 3$ projects, $H = 24$ periods, and $K = 5$ scenarios:

$$|\mathcal{S}| = 5^{3\times 24} = 5^{72} \approx 10^{50}, \tag{18}$$

already computationally intractable. At larger portfolio scales ($n = 10$, $H = 24$), the leaf count reaches $5^{240} \approx 10^{167}$. This exponential scaling is not a function of solver technology; it is intrinsic to the multi-stage stochastic formulation and grows multiplicatively with each additional project or planning period. The endogenous active-set structure (termination events change $\mathcal{P}_t^{\text{active}}$) adds a second source of combinatorial growth beyond pure scenario enumeration.

This intractability motivates replacing exhaustive enumeration with a sequential interaction framework: the portfolio manager is modelled as a policy that observes the current state, commits an allocation, and receives a reward signal, with the environment transitioning stochastically to the next state. This is precisely the Markov Decision Process formulation, and it forms the primary contribution of this paper.

3.5 Markov Decision Process Formulation

3.5.1 MDP Definition

Definition 1 (Portfolio Governance MDP)**.** The portfolio budget allocation problem is formalised as a finite-horizon Markov Decision Process (**?**):

$$\mathcal{M} = \big(\mathcal{S},\ \mathcal{A},\ \mathcal{T},\ R,\ \gamma,\ H\big), \tag{19}$$

with components defined below.

State space $\mathcal{S}$. The state at period t is the portfolio state S_t, which collects for each project i the following components:

- actual progress $P_{i,t}^{\text{actual}} \in [0, 1]$,
- cumulative actual cost $\text{ACWP}_{i,t} \in \mathbb{R}_{\geq 0}$,
- performance indices $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{TCPI}_{i,t}$, $\text{EAC}_{i,t}$,
- project status $\text{status}_{i,t} \in \{\texttt{pending}, \texttt{active}, \texttt{completed}, \texttt{terminated}\}$,
- cure-period counter $\tau_{i,t}^{\text{rem}} \in \{0, 1, \ldots, \tau_i^{\text{tol}}\}$,

together with the portfolio cash balance $B_t \in \mathbb{R}_{\geq 0}$. Formally:

$$S_t \in \mathcal{S} \subseteq [0,1]^n \times \mathbb{R}^n_{\geq 0} \times \mathbb{R}^{4n} \times \mathcal{Q}^n \times \mathbb{Z}^n_{\geq 0} \times \mathbb{R}_{\geq 0}, \tag{20}$$

where $\mathcal{Q} = \{\texttt{pending}, \texttt{active}, \texttt{completed}, \texttt{terminated}\}$. The state dimension is $9n + 1$ per period. All components are observable. The sole unobservable quantity is $\eta_{i,t}$, realised only after allocation $x_{i,t}$ is committed; it is not part of the state vector but drives the stochastic transition.

The Markov property holds exactly: S_{t+1} depends on $(S_t, \mathbf{x}_t, \boldsymbol{\eta}_t)$ only, not on the history $\mathcal{H}_{t-1} = (S_0, \mathbf{x}_0, \ldots, S_{t-1}, \mathbf{x}_{t-1})$. The full portfolio state S_t is a sufficient statistic for the history, making the MDP formulation exact rather than approximate.

Action space $\mathcal{A}$ and feasible set. The raw action space is $\mathcal{A} = \mathbb{R}^n_{\geq 0}$, with the state-dependent feasible set:

$$\mathcal{A}(S_t) = \left\{ \mathbf{x}_t \in \mathbb{R}^n_{\geq 0} \;\middle|\; \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad x_{i,t} = 0 \text{ if } \text{status}_{i,t} \neq \texttt{active} \right\}. \tag{21}$$

Action projection. To guarantee $\mathbf{x}_t \in \mathcal{A}(S_t)$ for any raw policy output $\tilde{\mathbf{x}}_t$, the environment applies a three-step projection before executing the transition:

(i) *Inactive zeroing.* Set $\tilde{x}_{i,t} \leftarrow 0$ for all i with $\text{status}_{i,t} \neq \texttt{active}$.

(ii) *Per-project allocation cap.* For each active project i, cap the allocation at the maximum that guarantees project completion in one period regardless of the efficiency draw:

$$\tilde{x}_{i,t} \leftarrow \min\left(\tilde{x}_{i,t},\ \frac{\text{BAC}_i}{\eta_i^{\min}}\right), \tag{22}$$

where $\eta_i^{\min}$ is the lower bound of the project's efficiency distribution (defaulting to 0.01 when no closed-form lower bound exists).

(iii) *Budget projection.* Normalise so the total allocation does not exceed the available balance:

$$\mathbf{x}_t = B_t \cdot \frac{\max(\tilde{\mathbf{x}}_t^{\text{active}}, \mathbf{0})}{\left\|\max(\tilde{\mathbf{x}}_t^{\text{active}}, \mathbf{0})\right\|_1 \vee 1}, \tag{23}$$

where $\tilde{\mathbf{x}}_t^{\text{active}}$ denotes the sub-vector restricted to active projects, and $\vee$ denotes the maximum operator ensuring the denominator is at least one when the agent outputs all-zero or all-negative values.

This projection is differentiable almost everywhere and preserves gradient flow through a policy network during training.

Transition kernel $\mathcal{T}$. The stochastic transition $\mathcal{T}(S_{t+1} \mid S_t, \mathbf{x}_t)$ is defined by the following sequence of operations executed within each period:

(i) *Early phase* (pre-allocation):

- Evaluate hard deadline (12); terminate immediately if triggered.
- Compute planned progress $P_{i,t}^{\text{plan}}$ via (2) and update EVM indices (6)–(9).

- Check milestone certifications: certify (i, j) if $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ and $t \geq e_{i,j}$; deliver $R_{i,j}^{\text{net}}$ via (4).

(ii) *Action application*: apply projection (23) to $\tilde{\mathbf{x}}_t$; draw $\eta_{i,t}$ from the project-specific distribution; update $P_{i,t}^{\text{actual}}$ via (1).

(iii) *Late phase* (post-allocation):

- Recompute EVM indices with post-allocation progress.
- Re-check certifications for milestones whose thresholds were crossed only after the allocation.
- Evaluate Condition 1 (11); update tolerance counter via (13).
- Execute soft termination if $\tau_{i,t}^{\text{rem}} = 0$.
- Compute termination settlement (14) and update V_t via (5).

(iv) *Balance update*: advance B_{t+1} via (15).

(v) *Status assignment*: set `completed` if $P_{i,t}^{\text{actual}} = 1$ and all milestones certified; advance $t_i^{\text{proj}} \leftarrow t_i^{\text{proj}} + 1$ for active projects.

3.5.2 Reward Function Design

The per-period reward is the discounted net cash flow:

$$r_t(S_t, \mathbf{x}_t, S_{t+1}) = \gamma^t \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right), \tag{24}$$

where V_t is the aggregate inflow (5) realised after the transition and $\sum_i x_{i,t}$ is the total allocation outflow. The policy optimisation problem is:

$$\pi^* = \text{argmax}_\pi \ \mathbb{E}_\pi \left[\sum_{t=0}^{H-1} r_t(S_t, \mathbf{x}_t, S_{t+1}) \right], \tag{25}$$

where the expectation is over the distribution of trajectories induced by π and the efficiency draws $\eta_{i,t}$.

Structural properties of the reward signal. The reward function (24) embeds all performance incentives structurally, without penalty terms or auxiliary objective components:

(i) *Completion incentive.* At project completion, $R_i^{\text{ret}} = \rho_i \cdot CP_i$ and the final milestone payment enter V_t simultaneously. A policy that allows a project to stall just below the final threshold foregoes this entire inflow, creating an incentive to cross all milestone thresholds.

(ii) *Termination deterrent.* A negative R_i^{term} reduces V_t at the termination period, directly depressing the reward. A positive R_i^{term} partially compensates but represents a project that has contributed less than its advance payment was worth. The termination mechanic thus acts as a structural penalty without any hand-crafted penalty coefficient.

(iii) *Timing incentive.* Under $\gamma < 1$, earlier certification of milestone j yields $R^{\text{net}}_{i,j}$ at a higher discount weight. A return-maximising policy will therefore front-load allocations to projects where earlier certification is feasible, without any additional objective term (Damodaran, 2012).

(iv) *No deviation penalty.* The reward contains no term that penalises deviations from a planned trajectory. All incentives emerge from the financial consequences of actual decisions.

Relationship to the upper bound. By Proposition 1, $\sum_t \mathbb{E}[r_t] \leq Z^*$ for any causal policy. The empirical gap $Z^* - \mathbb{E}[\sum_t r_t]$ is the performance quantity reported in Section 5 for each benchmark solution and configuration.

3.5.3 Observation Vector

The observation returned to a policy at each period is a flattened and normalised projection of S_t. Rather than exposing raw EVM indices, the environment computes seven actionable per-project signals that directly encode the decisions relevant to allocation:

$$o_t = \Big(\{ \tilde{\tau}^{\text{rem}}_{i,t},\ \tilde{x}^{\text{cu}}_{i,t},\ \tilde{x}^{\text{cu}+1}_{i,t},\ \Delta\tilde{\psi}^{*}_{i,t},\ \Delta\tilde{\tau}^{*}_{i,t},\ \tilde{x}^{*}_{i,t},\ \widetilde{\text{NPV}}^{*}_{i,t} \}_{i \in \mathcal{P}},\ \widetilde{\Delta B}_t,\ \tilde{B}_t \Big), \tag{26}$$

giving dimension $7n + 2$. The per-project components are described in Table 3. The portfolio-level components $\widetilde{\Delta B}_t$ and $\tilde{B}_t$ are the last-period net cash flow and the current available balance, both normalised by B_0. All components are observable at decision time.

Index	Symbol	Clip	Description
$7i$	$\tilde{\tau}^{\text{rem}}_{i,t}$	$[0, \tau^{\text{tol}}_i]$	Cure-period tolerance remaining
$7i+1$	$\tilde{x}^{\text{cu}}_{i,t}$	$[0,1]$	Minimum allocation to avoid a breach *this* period, normalised by B_0
$7i+2$	$\tilde{x}^{\text{cu}+1}_{i,t}$	$[0,1]$	Minimum allocation to avoid a breach *next* period, normalised by B_0
$7i+3$	$\Delta\tilde{\psi}^{*}_{i,t}$	$[0,1]$	Progress gap to next uncertified milestone
$7i+4$	$\Delta\tilde{\tau}^{*}_{i,t}$	$[-1,1]$	Timestep gap to next milestone's certification window, normalised by H
$7i+5$	$\tilde{x}^{*}_{i,t}$	$[0,1]$	Allocation required to reach next milestone target, normalised by B_0
$7i+6$	$\widetilde{\text{NPV}}^{*}_{i,t}$	$[-1,1]$	Discounted net payment of next milestone, normalised by $\max_i \lvert\text{NPV}^{*}_{i,t}\rvert$
$7n$	$\widetilde{\Delta B}_t$	$[-2,2]$	Last-period net cash flow, normalised by B_0
$7n+1$	$\tilde{B}_t$	$[0,2]$	Available budget, normalised by B_0

Table 3: Observation vector components (26). Per-project block i occupies indices $7i$ through $7i+6$. All normalisation denominators use $B_0 = \kappa \sum_i \text{BAC}_i$.

3.5.4 Gymnasium-Compatible Implementation

The MDP $\mathcal{M}$ is implemented as a configurable Gymnasium-compatible environment (Towers et al., 2024). The environment exposes the standard `reset()` / `step()` interface, with `observation_space` of dimension $7n+2$ and `action_space` $\mathcal{A} = [0,1]^n$ (raw fractions of the available budget). All project and portfolio parameters (Table 2) are passed as a configuration dictionary at initialisation; the planning horizon H and the number of projects n are derived automatically, making the environment fully parameterisable across contract types and portfolio sizes without code modification.

At each `step()` call the environment (i) applies the three-step projection (23), (ii) executes the transition kernel of Section 3.5.1, and (iii) returns the observation (26), the reward (24), the episode termination and truncation flags, and an `info` dictionary carrying the full per-project state for logging and analysis. Episodes terminate when $t \geq H$ or when all projects have reached `completed` or `terminated` status. The efficiency $\eta_{i,t}$ is drawn each period from the configured project-specific distribution; the distribution family and parameters are part of the configuration dictionary.

3.5.5 Benchmark Policies

Three deterministic benchmark policies are defined over the observation vector (26) to establish performance baselines and quantify the gap to the full-foresight upper bound. Each policy reads directly from the catchup and milestone fields and is normalised through the same projection (23) as any other agent.

Reach-Plan (π^{rp}). At each period, allocate exactly enough to bring every active project to its S-curve planned progress:

$$x_{i,t}^{\text{rp}} = \max\bigl(0,\ P_{i,t}^{\text{plan}} - P_{i,t}^{\text{actual}}\bigr) \cdot \text{BAC}_i. \tag{27}$$

This policy closes the progress gap entirely each period but does not account for milestone timing, contract profitability, or budget scarcity.

Catchup-T (π^{cu}). Allocate only enough to stay within the contractual progress delay cap $\delta_i^{\max}$ this period:

$$x_{i,t}^{\text{cu}} = \max\bigl(0,\ P_{i,t}^{\text{plan}} - \delta_i^{\max} - P_{i,t}^{\text{actual}}\bigr) \cdot \text{BAC}_i. \tag{28}$$

This is the minimum spend that avoids a breach flag this period.

Catchup-Next-T ($\pi^{\text{cu+1}}$). A one-period lookahead: allocate enough this period so that the projected delay *next* period also remains within $\delta_i^{\max}$:

$$x_{i,t}^{\text{cu+1}} = \max\bigl(0,\ P_{i,t+1}^{\text{plan}} - \delta_i^{\max} - P_{i,t}^{\text{actual}}\bigr) \cdot \text{BAC}_i, \tag{29}$$

where $P_{i,t+1}^{\text{plan}}$ is the planned progress at the next period, available from the S-curve (2). This avoids both current and next-period breaches at the cost of higher early expenditure and correspondingly lower discounted returns.

4 Evaluation

5 Benchmark Evaluation and Optimality Gap

This section reports the baseline benchmark study conducted to establish a performance floor, motivate the need for learned allocation policies, and quantify the improvement target available to a future reinforcement learning agent. Three deterministic greedy heuristics and a manually computed full-information optimum are evaluated across four benchmark portfolio configurations of increasing structural complexity. All episodes are executed through the Gymnasium-compatible environment described in Section 3, with the SQLite audit ledger providing a complete per-period record of allocations, cash-flow events, and breach flags for every episode.

5.1 Shared Parameters and Duration Convention

All four configurations share the contract and environment parameters listed in Table 4. These are fixed across every episode and every solver to ensure that observed performance differences are attributable solely to configuration structure and solver policy.

Parameter	Symbol	Value
Retention rate	ν_i	0.05
Progress delay cap	$\delta_i^{\max}$	0.15
Finish delay cap	$\Delta_i^{\max}$	2 periods
Termination tolerance	θ_i	2 breach-periods
Annual interest rate	r	0.12
Per-period deficit rate	$r/12$	0.01
Budget tightness	κ	3.0
Profit margin	π_i	0.20
Discount factor	γ	0.95
S-curve shape	(a_i, b_i)	(1.5, 3.0)

Table 4: Parameters shared across all benchmark configurations. Retention is deducted from each milestone gross payment after advance recovery has been applied; the accumulated retention is released in full at the final milestone.

Duration convention. A project with `planned_duration` $= D$ executes over project-timesteps $\tau_i \in \{0, 1, \ldots, D\}$, giving $D + 1$ active periods. The final planned timestep is $\tau_i = D$, at which the last planned milestone is due. With finish delay cap $\Delta_i^{\max} = 2$, the hard deadline falls at episode period $f_i + 2$, and the total efficiency sequence supplied to the environment has length $D + 1 + 2 = D + 3$ entries covering all possible active periods.

Deduction order. Within each milestone payment, advance recovery is applied first to the gross payment; the retention deduction is then applied to the same gross payment independently. At the final milestone, any advance not yet recovered is deducted in full, and all accumulated retention is released. This ordering — advance recovery before retention, both applied to gross — is validated consistently across all configurations.

5.2 Benchmark Configurations

Table 5 summarises the four benchmark configurations. Each is designed to isolate a specific structural mechanic of the environment, so that the failure mode of each greedy heuristic can be attributed to a concrete feature of the problem.

TC-10SF-R — Isolated atom. One project, one non-advance milestone, no advance payment. With $D_i = 0$, the project has a single planned action at $\tau_i = 0$, and the hard deadline is at episode period 2. The efficiency sequence is [0.91, 1.17, 0.83]. Contract price $P = 100 \times 1.20 = 120$. The single milestone carries weight $w = 1.0$, so gross payment $= 120$. Retention of $0.05 \times 120 = 6$ is withheld and immediately released at the same (final) milestone, giving net payment $= 120$. There is no advance payment

Config	N	BAC_i	D_i	s_i	M_i	ϕ_i	ψ_i^{trig}	**Primary mechanic**
TC-10SF-R	1	100	0	0	1	0.00	—	Isolated atom: single action, single milestone
TC-1LMAIF-R	1	100	3	2	3	0.20	0.80	Delayed start, progress-triggered advance, high reco
TC-10MAIF-R	1	100	3	0	3	0.20	0.00	Diminishing efficiency, termination risk
TC-3XMAIF-R	3	100/200/300	3	0/6/7	3	0.20	0.00	Multi-project, gap then overlap, interacting cash flo

Table 5: Benchmark configuration summary. N: number of projects; D_i: planned duration (so contractual active periods = D_i+1); s_i: planned start (episode period); M_i: number of non-advance milestones; ϕ_i: advance rate; ψ_i^{trig}: progress threshold at which the advance is disbursed.

or recovery. This configuration provides the cleanest possible baseline: it tests the fundamental progress-to-cash-flow loop with no confounding payment mechanics.

TC-1LMAIF-R — Delayed start and progress-triggered advance. One project, planned start $s_i = 2$, planned duration $D_i = 3$ (four active periods: $\tau_i \in \{0, 1, 2, 3\}$), three non-advance milestones, and an advance rate $\phi_i = 0.20$ with advance trigger $\psi_i^{\text{trig}} = 0.80$. The trigger means the advance payment of $0.20 \times 120 = 24$ is disbursed only once cumulative progress $EV_{i,t} \geq 0.80$ is reached — *not* at project start. Consequently the advance is absent from the first interim milestone and the associated deductions begin only after the trigger fires.

Advance recovery rate is $\rho_i = 0.50$, the highest of all configurations. With milestone gross payments of 18, 42, and 60 (weights 0.15, 0.35, 0.50), the recovery trace is: $j = 1$: $\min(0.50 \times 18,\ 24) = 9.0$ recovered, advance remaining = 15.0; $j = 2$: $\min(0.50 \times 42,\ 15) = 15.0$ recovered, advance fully cleared; $j = 3$: no further recovery. Retention deductions (0.9, 2.1, 3.0) accumulate across the three milestones and are released in full (6.0) at $j = 3$. Net payments after deductions are 8.1, 24.9, and 63.0. The efficiency sequence alternates: [0.91, 1.17, 0.83, 0.91, 1.17, 0.83]. This configuration validates: (i) delayed project start and its interaction with the episode clock; (ii) progress-triggered advance disbursement; (iii) high advance recovery rate and the liquidity trough it creates after the advance is received; and (iv) the ordering invariant that advance recovery is deducted before retention on every gross payment.

TC-10MAIF-R — Diminishing efficiency and termination risk. One project, planned start $s_i = 0$, planned duration $D_i = 3$, three milestones, advance rate $\phi_i = 0.20$ (trigger at progress = 0, i.e. disbursed at project start), advance recovery rate $\rho_i = 0.20$. The efficiency sequence is monotonically *decreasing*: [0.95, 0.91, 0.55, 0.40, 0.30, 0.20]. Early periods are productive; late periods are near-inoperable.

The advance of 24 is disbursed at $t = 0$. Milestone payment trace (advance recovery before retention): $j = 1$: $\min(0.20 \times 18,\ 24) = 3.6$ recovered, advance remaining = 20.4, retention = 0.9, net = 13.5; $j = 2$: $\min(0.20 \times 42,\ 20.4) = 8.4$ recovered, advance remaining = 12.0, retention = 2.1, net = 31.5; $j = 3$: $\min(0.20 \times 60,\ 12.0) = 12.0$ recovered (advance fully cleared at the final milestone via the standard rate), retention = 3.0, released = 6.0, net = 51.0. This configuration provides a balanced three-milestone structure while stress-testing the termination mechanic: the sharp efficiency drop beyond $\tau_i = 1$ makes it progressively more expensive to maintain progress above $\delta_i^{\max} = 0.15$, and any policy that does not front-load work risks triggering the cure-period counter in the later periods.

TC-3XMAIF-R — Multi-project with gap and overlap. Three simultaneous projects with heterogeneous scale (BAC = [100, 200, 300], prices [120, 240, 360]) and staggered starts ($s_i = [0, 6, 7]$). Each project has planned duration $D_i = 3$ (four active periods), three milestones, and advance rate $\phi_i = 0.20$ triggered at project start. Advances at episode periods 0, 6, and 7 are 24, 48, and 72 respectively, creating three distinct outflow spikes that the policy must navigate against a shared budget.

The episode timeline has two structurally distinct phases. *Gap phase* ($t = 4, 5$): Project 0 completes at $t = 3$; Projects 1 and 2 have not yet started. No allocations are required and the budget accumulates interest on any surplus. *Overlap phase* ($t = 7, 8, 9$): both Projects 1 and 2 are simultaneously active, creating competing allocation demands against the shared reserve B_t. The concurrent advance outflows at $t = 6$ (-48) and $t = 7$ (-72) produce the largest single-period deficit in the benchmark suite.

Per-project efficiency sequences are: Project 0: [0.91, 1.17, 0.83, 0.91, 1.17, 0.83]; Project 1: [1.09, 0.84, 1.18, 1.09, 0.84, 1.18]; Project 2: [0.97, 1.13, 0.86, 0.97, 1.13, 0.86]. None is monotonically degrading; the challenge here is purely one of coordination and cash-flow timing, not individual project difficulty.

5.3 Solvers

Four methods are evaluated, as specified in Section **??**.

Reach-Plan (π^{rp}) allocates $x_{i,t}^{\text{rp}}$ to bring every active project exactly to its S-curve planned progress each period, regardless of budget scarcity or milestone proximity.

Catchup-T (π^{cu}) allocates the minimum spend $x_{i,t}^{\text{cu}}$ required to keep the progress delay within $\delta_i^{\max} = 0.15$ this period, preserving budget for later periods and other projects.

Catchup-Next-T ($\pi^{\text{cu+1}}$) pre-emptively allocates $x_{i,t}^{\text{cu+1}}$ so that the projected delay next period also remains within $\delta_i^{\max}$, at the cost of higher early expenditure.

Full-information optimum (π^*) is computed manually with complete prior knowledge of all efficiency realisations ($\eta_{i,t}$). It serves as an upper-bound reference for each configuration. Implementing a tractable algorithmic solver — whether Linear Program or Dynamic Program — is left as future work; the manual computation suffices to establish the performance ceiling across the four configurations studied here.

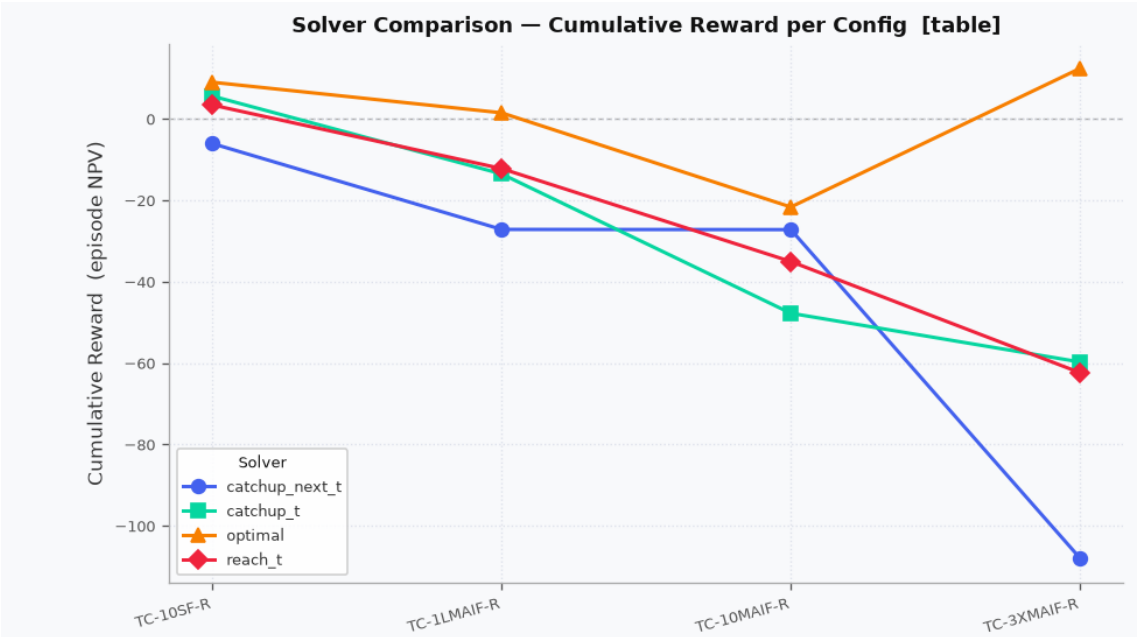
All methods are evaluated under seed = 0, with the fixed custom ordinal efficiency sequences described above, ensuring exact reproducibility.

5.4 Results

Figure 7 reports cumulative episode NPV for each method across the four configurations, ordered left to right by increasing portfolio complexity. Table 6 presents the numerical values and the best-heuristic gap $G(\pi^*) - \max_\pi G(\pi)$ for each configuration.

Config	$G(\pi^*)$	$G(\pi^{\text{rp}})$	$G(\pi^{\text{cu}})$	$G(\pi^{\text{cu+1}})$	**Best-heuristic gap**
TC-10SF-R	9.00	5.60	3.40	−6.07	3.40
TC-1LMAIF-R	1.49	−13.21	−13.21	−27.17	14.70
TC-10MAIF-R	−21.66	−35.06	−47.77	−27.22	5.56
TC-3XMAIF-R	12.40	−59.75	−62.38	−107.96	72.15

Table 6: Episode NPV G for all methods across four configurations. Best-heuristic gap $= G(\pi^*) - \max_{\pi \in \{\pi^{\text{rp}}, \pi^{\text{cu}}, \pi^{\text{cu+1}}\}} G(\pi)$.



	10SF-R	1LMAIF-R	10MAIF-R	3XMAIF-R
catchup_next_t	-6.07	-27.17	-27.22	-107.96
catchup_t	5.60	-13.53	-47.77	-59.75
optimal	9.00	1.49	-21.66	12.40
reach_t	3.40	-12.21	-35.06	-62.38

Figure 7: Cumulative episode NPV ($G = \sum_{t=0}^{T} \gamma^t \Delta B_t$) for three deterministic greedy heuristics and the full-information optimum across four benchmark configurations ordered by increasing portfolio complexity. The widening gap confirms that rule-based allocation deteriorates systematically as structural complexity grows.

TC-10SF-R. On the isolated single-period configuration all three heuristics produce positive NPV and the gap is modest. Reach-Plan achieves 5.60 against the optimal 9.00 (gap 3.40); Catchup-T follows at 3.40. Catchup-Next-T finishes at −6.07: with only one active period and one milestone, its one-period lookahead over-spends relative to what the single milestone requires, incurring interest costs without a compensating return. The result confirms that the environment correctly prices conservative over-allocation as a loss.

TC-1LMAIF-R. The progress-triggered advance and high recovery rate ($\rho_i = 0.50$) create a liquidity trough immediately after the advance is disbursed. All three heuristics produce negative NPV while the optimum remains marginally positive at 1.49; the best-heuristic gap is 14.70. The optimal allocation exploits the known efficiency sequence to time the milestone certifications so that the advance trigger fires at the most advantageous moment and the large recovery deductions at $j = 1$ (−9.0) and $j = 2$ (−15.0) are offset by subsequent cash inflows. The greedy policies, which allocate based on current-period signals only, cannot anticipate the trigger timing and over-allocate during the low-return period before the advance fires. The fact that both π^{rp} and π^{cu} achieve identical NPV (−13.21) indicates that the binding constraint here is not the delay cap but the payment timing: both rules happen to satisfy the tolerance in the same periods, but neither correctly sequences allocation to capture the advance inflow early.

TC-10MAIF-R. The monotonically diminishing efficiency sequence makes the allocation timing problem acute. Even the full-information optimum achieves only −21.66: the late-stage efficiency collapse ($\eta = 0.30$, 0.20 in the delay window) makes full project recovery uneconomical. All heuristics perform materially worse. Catchup-Next-T (−27.22) outperforms both Reach-Plan (−35.06) and Catchup-T (−47.77): its risk-averse lookahead front-loads work into the high-efficiency early periods, reducing expenditure in the near-inoperable late periods where each unit of allocation buys very little progress. Catchup-T performs worst because its minimum-spend rule continues allocating into low-efficiency periods to maintain the tolerance floor, accumulating cost without meaningful progress gain. The best-heuristic gap of 5.56 is the smallest in the benchmark suite — but even here the optimum is materially better and the gap arises from information the greedy rules cannot access.

TC-3XMAIF-R. The multi-project configuration produces the starkest result. The full-information optimum achieves +12.40: coordinating across three projects with known efficiency sequences, the oracle times advance disbursements, milestone certifications, and the gap-phase budget accumulation to convert the complex portfolio into a profitable episode. Every greedy heuristic produces a loss: Reach-Plan at −59.75, Catchup-T at −62.38, and Catchup-Next-T at −107.96. The best-heuristic gap is 72.15, a 21× increase over TC-10SF-R. The sign reversal — profitable optimum, loss-making heuristics — is the critical observation. The concurrent advance outflows at $t = 6$ and $t = 7$ (−48 and −72) create a large shared deficit that each single-project greedy rule handles independently, without accounting for how its own allocation depletes the reserve available to the concurrent project. Catchup-Next-T collapses to −107.96 because its aggressive pre-emptive spend across all three simultaneously-active projects exhausts the budget during the overlap phase, leaving insufficient reserve to certify the high-value late milestones of Projects 1 and 2.

5.5 Discussion

Three findings emerge from the benchmark study.

The gap is monotonically increasing with structural complexity. The best-heuristic gap grows from 3.40 (TC-10SF-R) to 14.70 (TC-1LMAIF-R) to 72.15 (TC-3XMAIF-R), a 21× increase from simplest to most complex. TC-10MAIF-R is the only exception to the monotone trend: its gap of 5.56 is smaller than TC-1LMAIF-R's 14.70 because the challenge there is project-level efficiency timing, not multi-project coordination. The gap in TC-3XMAIF-R is not marginal degradation; it reflects a structural failure of greedy allocation under shared-budget coordination requirements that no single-project rule can resolve.

No single heuristic dominates. Each heuristic is best on at least one configuration. Reach-Plan is best on TC-10SF-R; Catchup-Next-T is best on TC-10MAIF-R; Catchup-T is best on TC-3XMAIF-R. This confirms that the optimal policy is adaptive: the correct allocation depends jointly on the efficiency trajectory, the milestone payment structure, the advance trigger timing, and the number and phasing of concurrent projects — a combination no fixed rule can capture.

A well-defined improvement target exists for learned agents. The full-information optimum $G(\pi^*)$ provides a concrete upper bound. On TC-3XMAIF-R the swing between the best achievable (12.40) and the best greedy (-59.75) is 72.15 NPV units, corresponding to the difference between a profitable and a deeply loss-making portfolio outcome. This gap is the value of intelligent sequential allocation — and it is precisely the target that a policy trained on the CMDP environment of Section 3 is positioned to close. Training such a policy and evaluating it against a formal Linear Programming bound constitute the natural next step, addressed in subsequent work.

6 Conclusion

6.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

(1) Literature-Calibrated Synthetic Data Generation: We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

(2) Rolling Horizon Control Architecture: We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

(3) Formal Problem Formulation: We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

(4) Complete Modeling Framework: We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with

all assumptions explicitly stated and future extensions identified.

6.2 Implications for Research and Practice

For Operations Research: This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

For Project Management: The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

For Machine Learning: This application showcases RL's potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

6.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

6.4 Future Work

Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)
2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas → construction → infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

6.5 Broader Impact

This research has potential impact beyond project portfolio management:

Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

6.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

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A Appendix A : PRISMA 2020 Flow Diagram

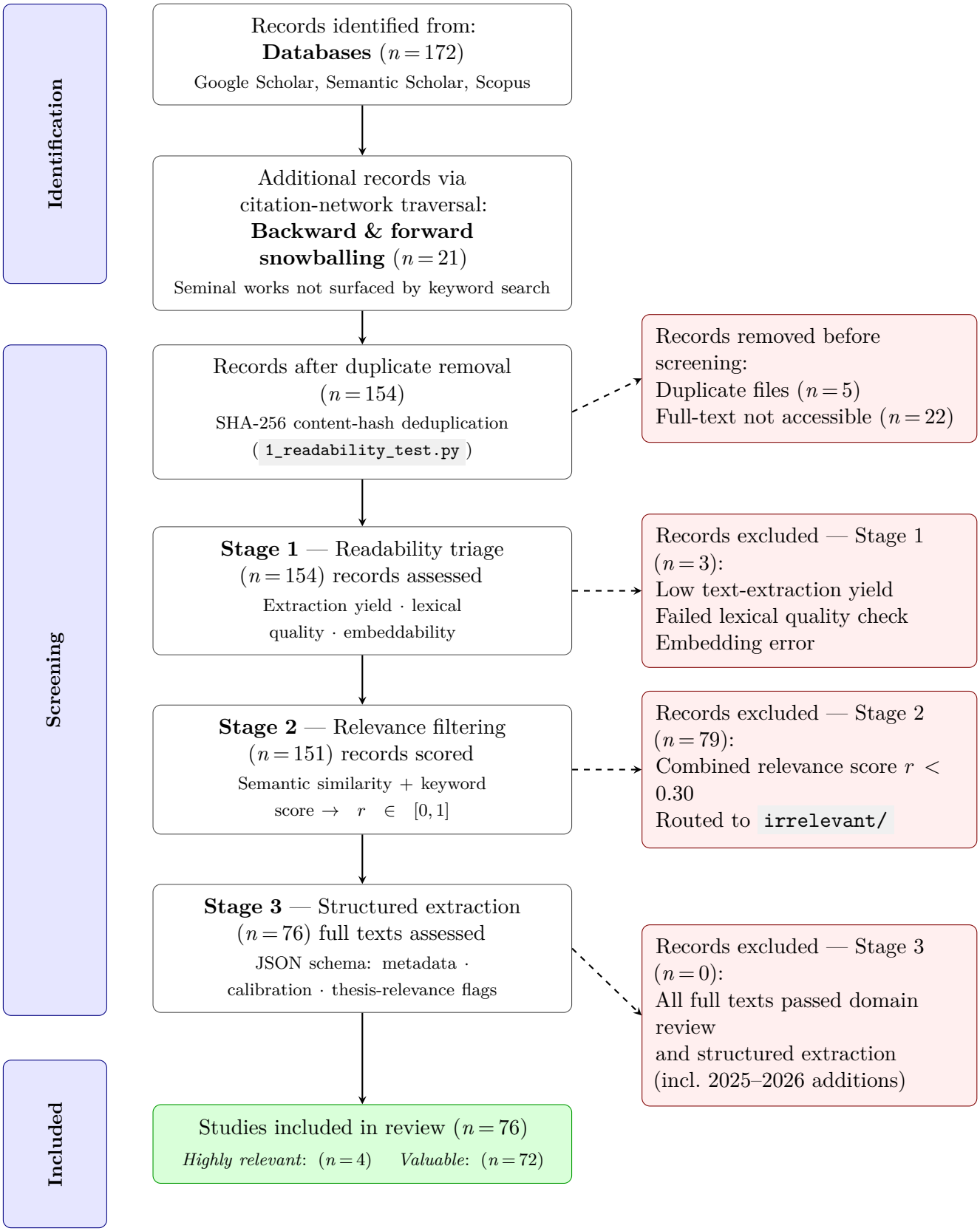


Figure 8: PRISMA 2020 flow diagram of the automated literature search and screening process. Stage labels correspond to scripts in `docs/lit_archive/`.

B Appendix B : Gymnasium Compatibility Certificate

`tests/gym_test.py` exercises `PortfolioBudgetingEnv` across ten test sections and 51 individual checks, covering import and instantiation, space declarations, the `reset()` and `step()` API contracts, determinism under fixed and varying seeds, random-policy and degenerate (zero-action, greedy) rollouts, multi-episode stability, and the official `gymnasium.utils.check_env` checker. The suite is run against benchmark configuration `TC-3XMAIF-R` (3 projects, mixed start times, fixed efficiency sequences). Table 7 summarises the ten sections; all 51 checks pass.

Table 7: Gymnasium compatibility test summary — `PortfolioBudgetingEnv`, config `TC-3XMAIF-R`, generated 2026-10-20.

§	Section	Passed
1	Import & Instantiation	8 / 8
2	Space Declarations	7 / 7
3	`reset()` Contract	13 / 13
4	`step()` Contract	10 / 10
5	Determinism	2 / 2
6	Random Policy Rollout	6 / 6
7	Zero-Action Rollout	1 / 1
8	Greedy Rollout	1 / 1
9	Multi-Episode Stability	2 / 2
10	`gymnasium.utils.env_checker`	1 / 1
Total		**51 / 51**

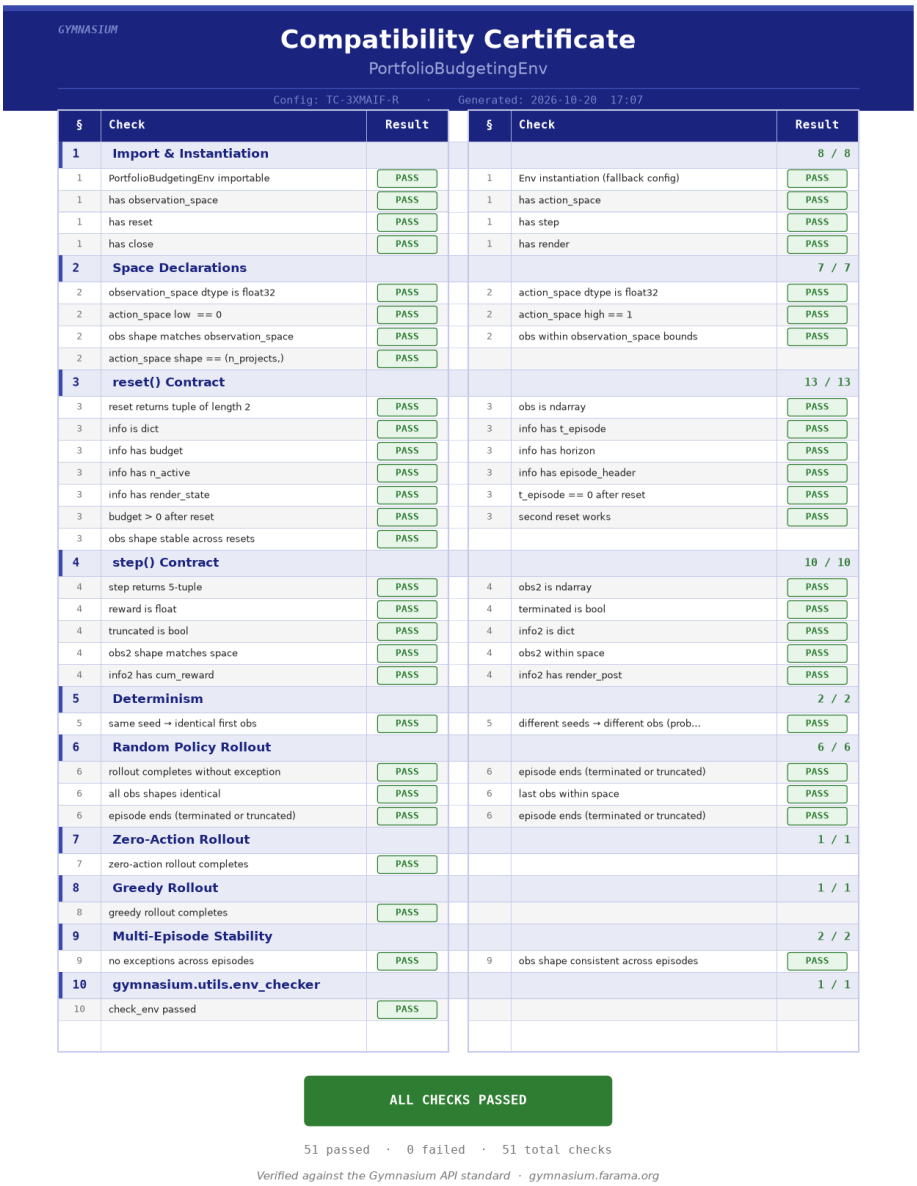


Figure 9: Compatibility certificate generated by `tests/gym_test.py`. All 51 checks pass; 0 failures.

C Reopsitory Architecture and Concept Map

C.1 Repository Architecture Overview

The complete implementation accompanying this paper — the environment, the baseline solvers, the verification suite, and the literature-review pipeline described in Section 2 — is organised as a single version-controlled repository. The repository separates *documentation* (this manuscript, the thesis, and the literature corpus) from *implementation* (the environment, ledger, baselines, and tests), as summarised below.

```
ppm_env/
|-- notation.yaml        parameter dictionary (academic <-> code)
|-- docs/
|   |-- article/          LaTeX source of this manuscript
|   |-- lit_archive/      literature screening pipeline and corpus
|   |-- mscn/             supplementary thesis documentation
|   |-- pp_presentation/  thesis defense slides (university deliverable)
|   `-- thesis_farsi/     Persian thesis manuscript (university deliverable)
`-- src/
    |-- db/               ledger schema, configs, run records
    |-- env/              core CMDP environment implementation
    `-- tests/            verification suite, baselines, gap analysis
```

The `mscn/`, `pp_presentation/`, and `thesis_farsi/` directories satisfy institutional degree-completion requirements and fall outside the reproducibility scope of this paper; they are listed here only for completeness. The `notation.yaml` file at the repository root provides an explicit, machine-readable correspondence between every academic symbol introduced in Section 3 and its counterpart variable name in the source code, and is intended as the primary cross-reference for readers wishing to connect the formal model to the implementation.

The literature-review pipeline described in Section 2.1.1 resides in `docs/lit_archive/` and retains, alongside the processing scripts, the full PDF archive (`lit_archive_passed/`, `lit_archive_failed/`) and the structured JSON extraction records that drove the synthesis table and gap-analysis figures reported in Section 2.

C.2 Core Contribution Modules

The modules implementing the formal model of Section 3 and the evaluation of Section 4 are contained entirely within `src/`. The environment dynamics — state construction, EVM/S-curve computation, the FIDIC cash-flow cycle, and the termination gate — are implemented in `src/env/`, with the environment's public Gymnasium-compatible interface and its internal support routines maintained as separate modules within that directory. During training or evaluation, the environment class is instantiated per episode through the standard Gymnasium `reset()` / `step()` interface; no additional driver component is required.

The SQLite audit ledger introduced in Section 3 is schema-defined and persisted under `src/db/`, which also holds the benchmark portfolio configurations used in Section 4 and the run-record databases generated by each executed episode.

The verification and evaluation layer resides in `src/tests/`, which bundles four functionally distinct components: the Gymnasium compliance suite, the baseline and optimal solvers, the benchmark configuration generator together with its archived reference episodes, and the post-hoc analysis scripts used to produce the figures and tables of Section 4.

```
src/tests/
|-- gym_test.py                 Gymnasium API compliance suite
|-- gym_certificate.png         compliance certificate (Appendix B)
|-- baseline_solvers.py         greedy heuristics and optimum solver
|-- reference_episodes_gen/     benchmark configuration generator
|-- reference_episodes_record/  archived configs and solutions
`-- analysis/                   gap-analysis and figure scripts
```

The `reference_episodes_record/` directory retains the configuration set and recorded solutions used to validate the environment and to produce the four benchmark configurations evaluated in Section 4; it is intended as a working template for practitioners who wish to instantiate the environment against their own company data.

C.3 Concept-to-Implementation Traceability Matrix

Table 8 maps every concept and promised contribution named in the main text to its corresponding location in the repository, and is provided to support independent verification of the claims made throughout this paper.

Table 8: Traceability matrix mapping concepts and promised contributions to their implementation in the repository.

Concept / Contribution	Ref.	Repository Location	Role
Academic–code notation dictionary	§3	`notation.yaml`	Maps formal symbols to code identifiers
EVM/S-curve state representation	§3	`src/env/`	State vector construction
FIDIC cash-flow transition dynamics	§3	`src/env/`	Advance, retention, interest, and milestone logic
Two-condition termination gate	§3	`src/env/`	Episode termination evaluation
SQLite audit ledger	§3	`src/db/`	Schema definition and per-run event logging
Benchmark portfolio configurations	§4	`src/db/`	Stored configuration instances for evaluation runs
Gymnasium compliance verification	App. B	`src/tests/`	API conformance tests and certificate generation
Greedy heuristic baselines (Reach-Plan, Catchup-T, Catchup-Next-T)	§4	`src/tests/`	Deterministic baseline policies

continued on next page

Table 8 – continued

Concept / Contribution	Ref.	Repository Location	Role
Manually computed full-information optimum	§4	`src/tests/`	Upper-bound benchmark solver
Benchmark configuration generation and reference episodes	§4	`src/tests/`	Configuration matrix, episode archive, and company-data templates
Gap-analysis figures and dashboards	§4	`src/tests/`	Post-hoc analysis and figure generation
Literature screening, Stages 1–3	§2.1.1	`docs/lit_archive/`	Readability triage, relevance filtering, structured extraction
Synthesis dashboard and literature map	§2.1.1	`docs/lit_archive/`	Gap matrix, domain charts, synthesis table generation
Manuscript source	—	`docs/article/`	LaTeX source, figures, and tables for this paper

C.4 Reproducibility Notes

The mapping in Table 8 reflects a specific snapshot of the repository at the time of writing; subsequent development may reorganise individual modules without affecting the directory-level structure described in this appendix. Access to the repository, including the full commit history, the benchmark configurations, and the Gymnasium compliance certificate reproduced in Appendix B, is available from the corresponding author upon request.